

mlx2

The capability book

A clean-generation inference runtime for Apple silicon that connects model-specific acceleration to exact state ownership, bounded memory and one shared API lifecycle. This edition covers a week of features, a ten-lane bug sweep and its review-fix round, before-and-after GPU campaigns through the real HTTP serving path, the first equal-heat thermal map, and a power governor a grid operator can turn down.

49 → 107

tok/s, Qwen3.8-27B native MTP at four lanes, baseline to final

9.5 → 0.47

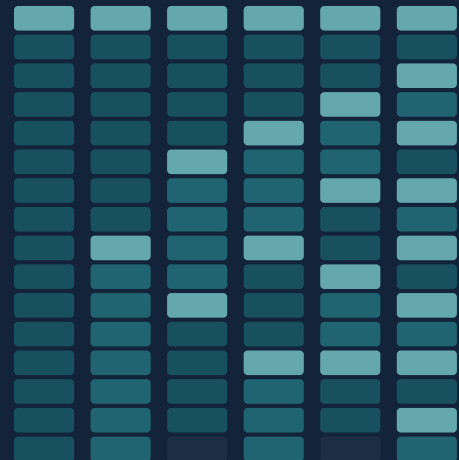
s, worst Flash-Next decode stall beside a 16K prompt

789 → 508

ms, worst p95 neighbour gap across the four 27B routes

24.5 W

average GPU + DRAM draw under a 25 W runtime budget



BASELINE

origin/main 187c6aa23 · every "before" number ran here

FINAL TREE

main @ 43d561e38 · 137 code commits after the baseline

PUBLISHED

GitHub pierre427/mlx2 8359344 (13 signed topic commits) · private main 43d561e38

EVIDENCE

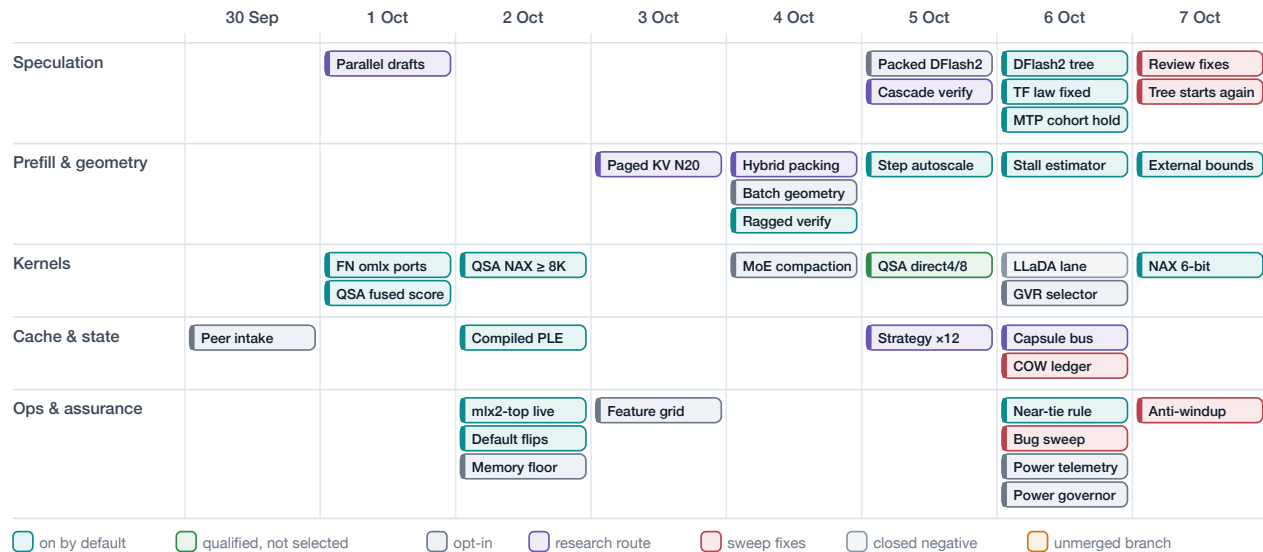
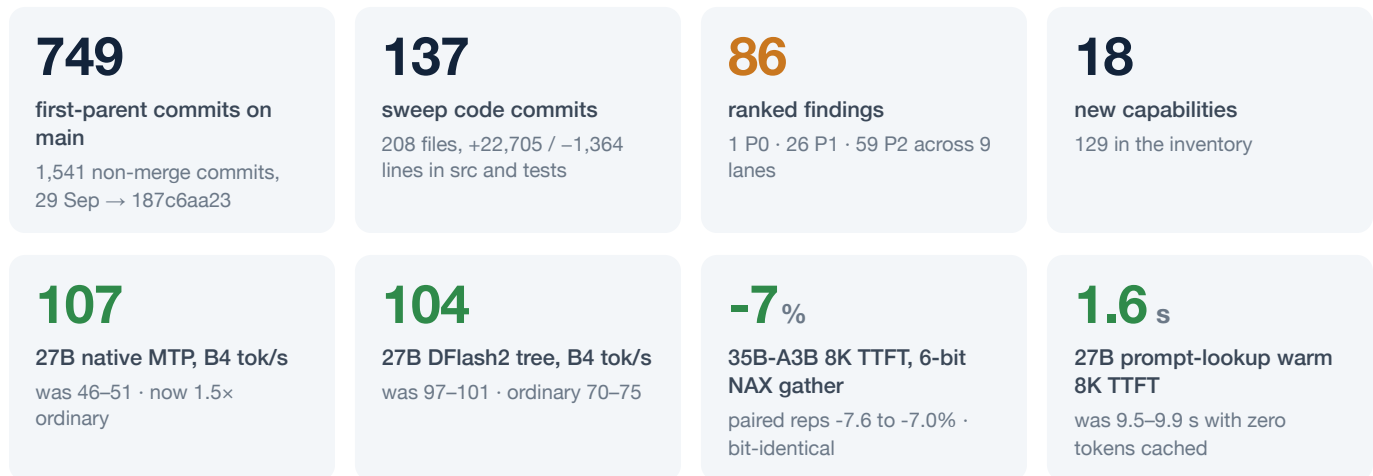
16 GPU campaigns in qualification/runs/sweep-20261006 · lane reports · research-20261006 · git

BUILT

2026-10-07 by docs/brag-book/build.py · experimental research software

Seven days, one honest scoreboard

A new default decode route for the flagship dense model, a sweep that read the whole tree and fixed what it found, and the GPU campaigns that measured the result before and after.



The week, by area and day. Each pill is a feature or change that landed (7 October is the review-fix and integration round). Colour is its status today, not its ambition. Source: `qualification/runs/sweep-20261006/lanes/history.md §1`; `git log`

HOW TO READ THE TAGS

ON BY DEFAULT selected without a flag · **QUALIFIED** matching receipt for source, runtime, artifact and settings · **OBSERVED-USED** counters prove it ran · **OPT-IN** needs a flag · **RESEARCH** default-off route under study · **UNQUALIFIED** runs, labelled, without that receipt. These are different states; a default is not a qualification.

SCOPE OF THE NUMBERS

16 GPU campaigns on one M5 Max laptop through the real HTTP path. **Before** ran on 187c6aa23; **after** ran on the final integration tree with every fix; same-code A/Bs (estimator, keep-warm, telemetry, governor modes) compare arms on one tree. Two repetitions in rotated order (three for the 35B), cooldowns between arms. A/B evidence for this host, not qualification and not a controlled performance run.

The same campaigns, on the baseline and on the final tree

Every chart in this chapter pairs a grey baseline bar with a coloured final bar, from the same workload, harness and host. The point is the arrow between them.

1.5×

27B native MTP over ordinary, B4

104–111 vs 70–75 tok/s;
baseline 0.73×

475 ms

Flash-Next worst stall beside 16K

was 6.9–9.5 s on the served policy

508 ms

worst p95 neighbour gap, 27B

all four routes; baseline worst 789 ms

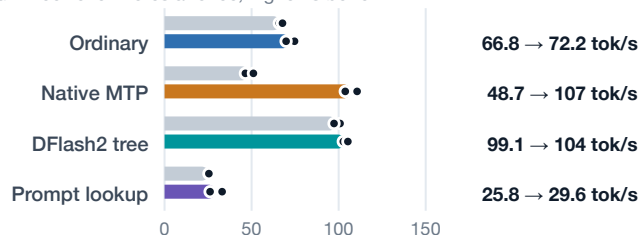
3.3 s

Flash-Next B8 median TTFT

was 6.8–7.0 s at the same batch wall time

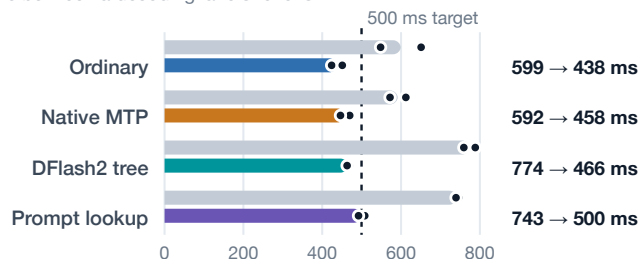
B4 aggregate decode

four ~700-token notes at once; higher is better



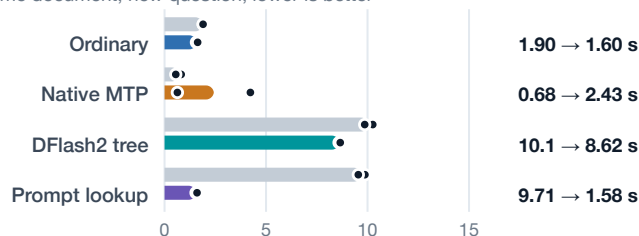
Neighbour gap p95, 12K prompt beside decode

time between a decoding lane's tokens



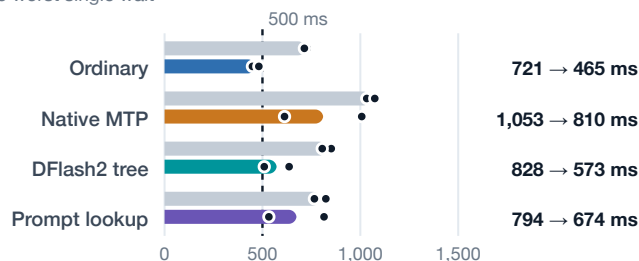
Warm 8K prefix: time to first token

same document, new question; lower is better



Neighbour gap max, same run

the worst single wait



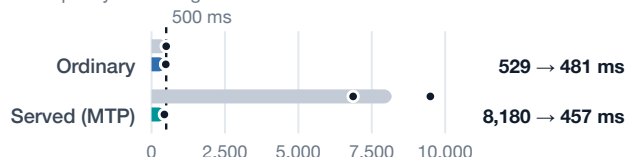
Qwen3.8-27B (oQ4e), four routes, baseline against the final tree. ■ baseline 187c6aa23 ■ final integration tree (each route's own colour). Bars are medians, dots repetitions. Native MTP's B4 rate more than doubled; every route's p95 neighbour gap reached the 500 ms target; prompt lookup's warm prefix became a cache hit. DFlash2's warm miss and the worst single gaps did not move enough (see the limits chapter). Source: [qualification/runs/sweep-20261006/q38-routes/results.json](#); [qualification/runs/sweep-20261006/final-q38-routes/results.json](#)

- **Native MTP B4** 46–51 → 104–111 tok/s: co-arrived prompts now form one MTP cohort instead of one MTP lane and three plain ones (SS-1). [56fce148e](#); [e4213207f](#)
- **Prompt lookup** warm 8K TTFT 9.5–9.9 → 1.55–1.60 s with 6,144 tokens cached; p95 739–746 → 492–508 ms. [cb26c206b](#); [673aa9fe0](#)
- **DFlash2 B4** 97–101 → 103–105 tok/s, and its neighbour p95 759–789 → 463–469 ms once the external loop got the stall bound. [673aa9fe0](#)
- **Prompt lookup B4** 25–26 → 55–63 tok/s on the closing run (ordinary 62–63): plain rounds now pipeline like ordinary decode. [82bdcc230](#)

- **Worst single gap** on native MTP is still 613–1007 ms; p95 meets the target, the tail does not.
- **Native-MTP warm TTFT** 0.55–0.82 → 0.64–4.23 s: one final repetition resumed from only 4,096 cached tokens. Hit depth is unstable (open).

Neighbour gap max, 16K prompt beside decode

the served policy's stall is gone



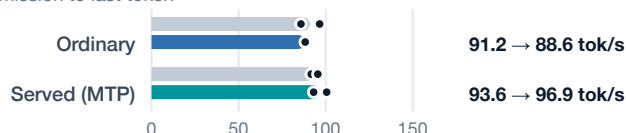
B8: median time to first token

eight co-arrived prompts; lower is better

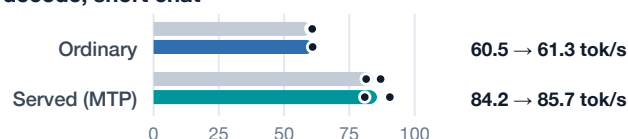


B8: tokens over batch wall time

submission to last token



B1 decode, short chat

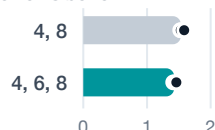


Flash-Next (Uncensored 4-bit MTP): the served policy against ordinary, before and after.

■ baseline 187c6aa23 ■ final integration tree. The served policy used to stall a decoding neighbour for 6.9–9.5 s beside a 16K prompt; on the final tree its worst gap is 440–475 ms. B8 is measured over batch wall time, submission to last token: the same 17.8 s and 93–101 tok/s, with the median first token twice as early. Ordinary's B8 median TTFT rose (4.2–4.9 → 5.3–5.8 s); it is not explained yet. Source: qualification/runs/sweep-20261006/fn/results.json; qualification/runs/sweep-20261006/final-fn/results.json

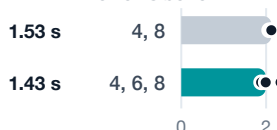
8K prompt TTFT

lower is better



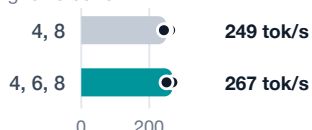
16K prompt TTFT

lower is better



B8 aggregate

higher is better



B1 decode

no gather in decode



Qwen3.6-35B-A3B Heretic: 6-bit experts join the fused NAX gather. Half of this model's layers carry 6-bit experts, which the stock gather handled at 21.5% of prefill. Gather widths admitted to NAX: grey the old set, teal the new default; three repetitions per arm. B1 decode is the control: it does not use the gather. Source: qualification/runs/sweep-20261006/final-q36-6bit/summary.json

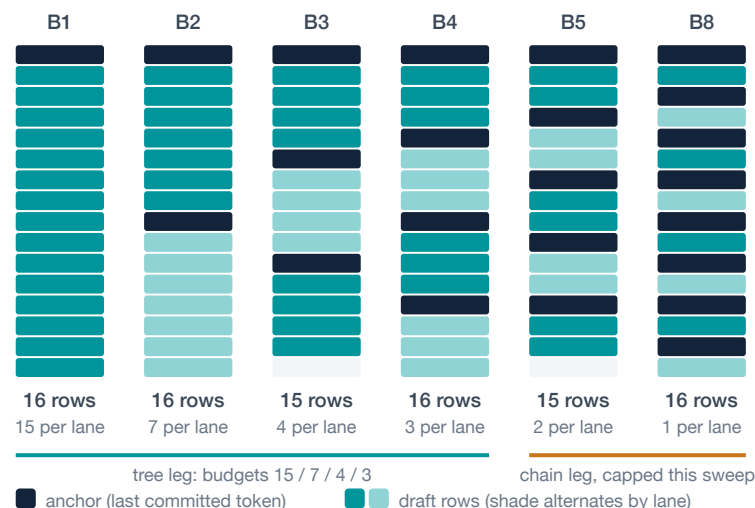
- 8K TTFT 1.52–1.58 → 1.41–1.46 s, paired repetitions -7.0 to -7.6%; 16K -6.3 to -9.2%; B8 +3.1 to +7.4%.
- Bit-identical: 102 canary variants exact; logits and 24 greedy tokens identical on the 35B. FINDINGS-GPU.md (nax-6bit)
- Default widths are now 4, 6, 8; MLX2_MOE_NAX_GATHER_BITS=4,8 restores the old set. Requalification advised.

MECHANISM 1 · SPECULATION

Qwen3.8-27B decodes with a DFlash2 token tree by default

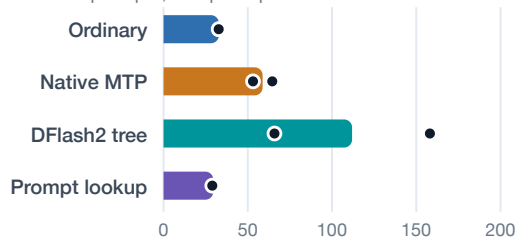
CANDIDATE, NOT QUALIFIED

1 · Drafter proposes a token tree (B1)

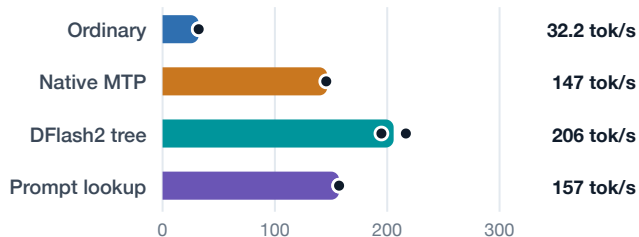


One verify forward, about sixteen rows, at every width. Tree budgets 15 / 7 / 4 / 3 for one to four lanes; above four lanes the route drafts a linear chain, which the sweep capped to the same row budget (2 drafts per lane at B5, 1 at B6–B8, none from B9). Source: [qualification/policies/qwen38-27b-dflash2-tensorfold-unified.json](#); 1f9c22d1d

two different prompts, one per rep



repetitive text: drafts land



5

- B1 short chat: DFlash2 66–158 tok/s across the two prompts against ordinary 32.7–33.0; the spread is the drafter's acceptance on different text. Tool-call TTFT 0.58–0.59 s against 1.43–1.51 s; JSON 59–60 against 31 tok/s.
- The tree gains nothing on decode after an 8K prompt: DFlash2 26–36 tok/s against ordinary 31, while native MTP holds 41–51.
- The week corrected the TensorFold numerical law and replaced it with an mlx2-owned tree GDN; a reversed BAAB run kept 420 outputs identical. [qualification/runs/sweep-20261006/lanes/history.md §1B](#)
- Still a candidate: the semantic gate fails (code fences, creative length) and the native-MTP handoff gate at width 8 missed by 9.8%. [qualification/runs/sweep-20261006/lanes/history.md §1D](#)

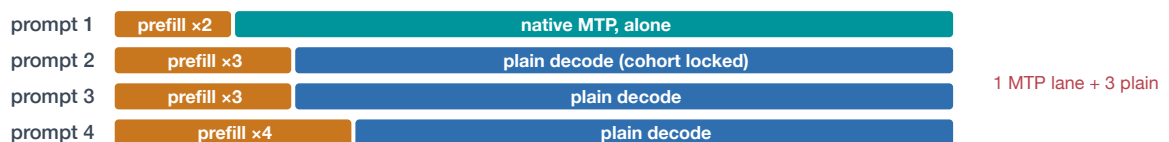
MECHANISM 2 · SCHEDULING · FIXED THIS SWEEP

Prompts that arrive together decode together

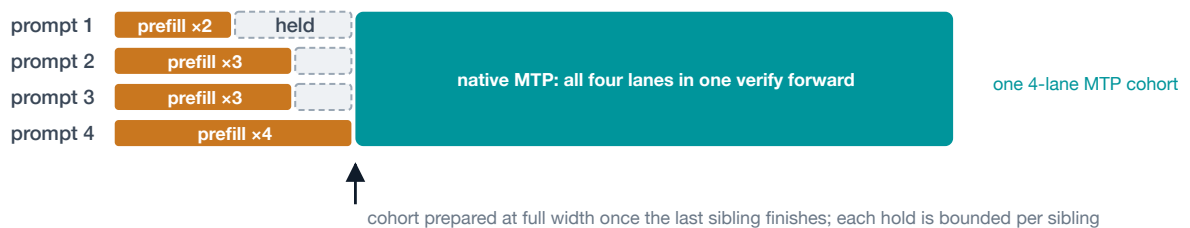
BEHAVIOUR CHANGE SS-1 MEASURED ON GPU

Self-MTP decodes a cohort of lanes in one verify forward, and the cohort's width is fixed when it starts. When four chat prompts arrived together and each needed more than one prefill chunk, the first to finish started MTP alone and the other three were left to decode plainly beside it.

Before (SS-1): four chat prompts arrive together



After: a bounded hold, then one cohort



A bounded hold turns one MTP lane and three plain lanes into one four-lane cohort. The first version bounded the hold by the siblings' summed prefill, which on Flash-Next's eight lanes released the prompts in waves; the hold is now bounded per sibling, and partially prefilled siblings are prepared together. Source: 56fce148e; e4213207f; 057373692

- Qwen3.8-27B native MTP at B4: 46–51 → 104–111 tok/s, now 1.5× ordinary (70–75). [q38-routes](#); [final-q38-routes](#)
- Flash-Next B8: median TTFT 6.8–7.0 → 3.0–3.6 s at the same batch wall time. [fn](#); [final-fn](#)
- The first finisher's TTFT now includes its siblings' bounded prefill; counter `mtp_coarrival_holds` shows when that happened.

A completion-time cost model bounds every prefill stall

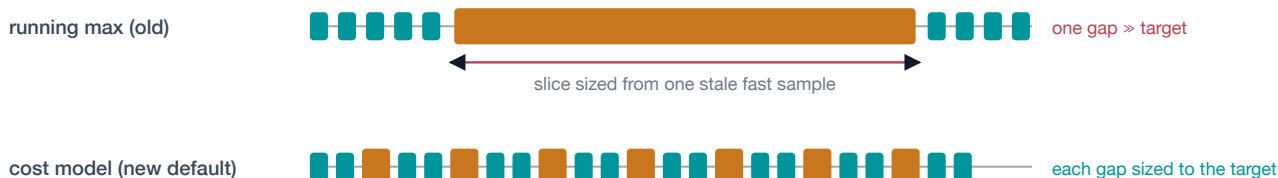
NEW DEFAULT THIS SWEEP

SAME-CODE GPU A/B

REQUALIFICATION ADVISED

Decode fairness lets long prompts prefill in slices beside live decode. The old bound kept a running maximum of per-row speed: one fast sample pinned it after a slowdown, a four-row forward cost four times the target, and self-MTP preparation never reported its time. The new estimator predicts each forward's completion time.

A decoding lane (teal) shares the GPU with a long prompt's prefill slices (amber)

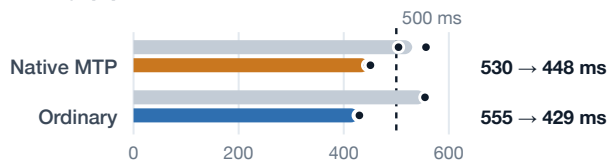


$t(R) = a + b \cdot R$ for each (route kind, KV-depth bucket)

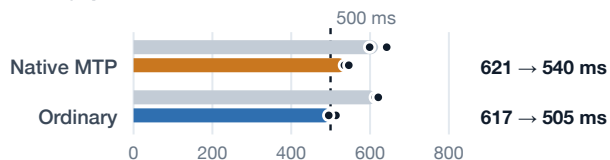
a: recent decode-step cost, clamped to half the cheapest prefill · b: 90th percentile of a 16-sample ring · R: rows the forward really runs
shift up fast on overrun ($r > 1.5$ twice, or > 3 once), down slowly ($4 \times r < 0.67$) · capped at $2 \times$ the largest forward since the last shift · 10% margin

From a stale maximum to a prediction. The cost model is kept per route kind and KV depth, counts the rows a forward really runs, and includes self-MTP preparation, which now owes decode debt. `decode_fairness.estimator: running_max` restores the old rule for A/B. Source: d28a774aa

Neighbour gap p95

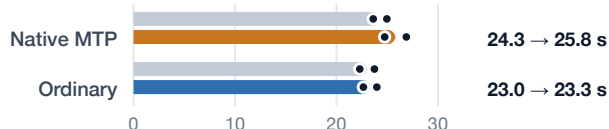


Neighbour gap max



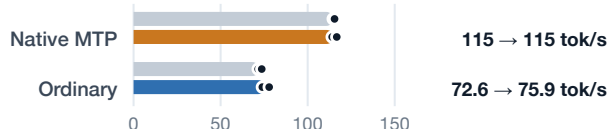
The long prompt's own TTFT

the price: slower long prefill



B4 aggregate decode

unchanged



Same code, two estimators. ■ running max (old rule) ■ cost model. Neighbour p95 -23 to -16%, worst gap -18 to -13%; the long prompt pays +1 to +6% on its own TTFT; B4 decode is unchanged. Source: qualification/runs/sweep-20261006/estimator/results.json

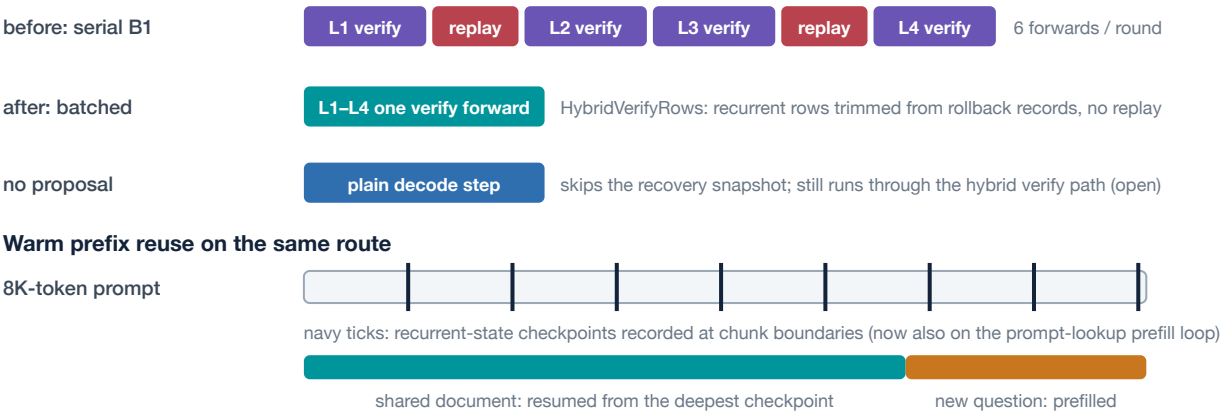
- The external-draft and prompt-lookup routes ran their own prefill loops with fairness off; they now apply the same bound (673aa9fe0), which is what brought DFlash2 and PLD under target in chapter 1.
- The estimator is recorded in route identity (acadd8b76), so receipts say which rule sized the slices.

Prompt lookup batches on recurrent hybrids and reuses warm prefixes

OPT-IN ROUTE SWEEP FIX B4 AT PARITY

Prompt lookup drafts by copying text the conversation already contains, which is free when the model repeats itself. On Qwen3.8's recurrent GDN layers it could not batch, and its prefill recorded no recurrent checkpoints, so a shared document was always a full miss.

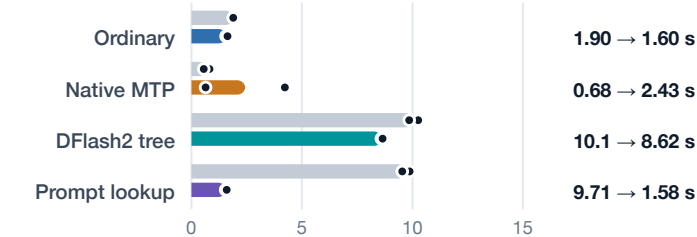
Prompt lookup at four lanes on Qwen3.8-27B (recurrent GDN layers beside KV attention)



Before and after on the GPU's timeline. Hybrid lanes share the external route's HybridVerifyRows transaction: one forward per round, recurrent rows trimmed from rollback records instead of replayed. Warm prefixes now resume from recorded checkpoints. Source: 2942ee30b; cb26c206b; 1c45a38a7

Warm 8K prefix: time to first token

lower is better



- Warm 8K TTFT 9.5–9.9 → 1.55–1.60 s; copy decode 157 tok/s against ordinary 32.
- B4 25–26 → 55–63 tok/s on the closing run (ordinary 62–63): the cohort polls every verified token, keeps its batched views and pipelines plain rounds like ordinary decode. FINDINGS-GPU.md (final-pld)

Warm-prefix TTFT, baseline against final. Prompt lookup now resumes from a checkpoint like ordinary. Native MTP's final median is pulled up by one repetition that hit a shallow checkpoint (open). DFlash2 still re-prefills by design: the external route pairs draft state only at the prompt boundary. Source: qualification/runs/sweep-20261006/final-q38-routes/results.json

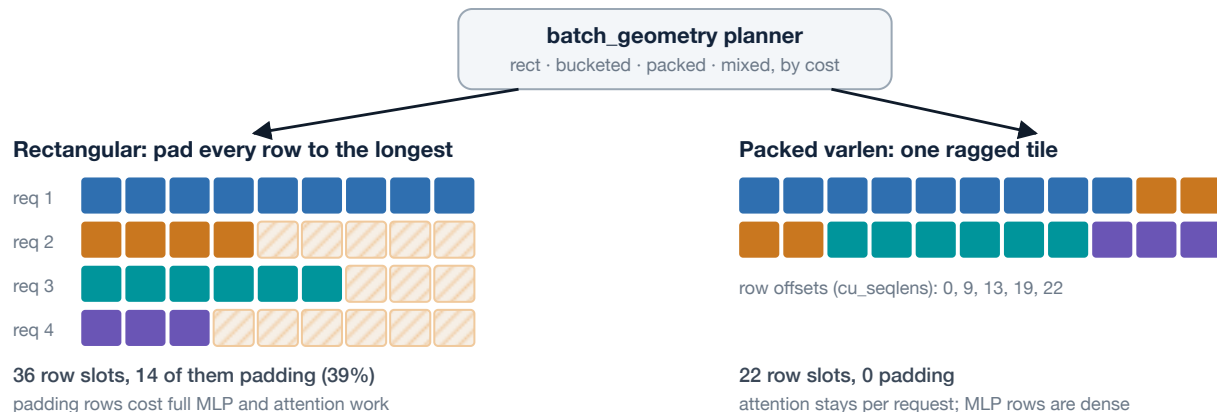
Packed variable-length rows, chosen by an adaptive geometry planner

RESEARCH ROUTE

DEFAULT-OFF

OBSERVED-USED

A rectangular batch pads every request to the longest one, and padding rows pay full MLP and attention cost. Packing concatenates the real rows into one ragged tile with per-request offsets; a `batch_geometry` planner chooses rectangular, bucketed, packed or mixed by cost.



What packing removes. Four requests of 9, 4, 6 and 3 tokens: the rectangular tile carries 14 padding slots out of 36; the packed tile carries none.



- Adaptive multi-user geometry replay avoided 17.45% of padded rows (a geometry measure, not speed).
- A 10-minute arbitrary-traffic A/B of the native N20 route was 7.44% slower overall; padding share alone did not predict wins.
- Sweep: compaction declines are counted (6b5f6aed3), the planner fails closed beside SRPT (4a0c00bae), the varlen pair is one operator-owned default (29056b2fb).

The win scales with padding. Native N3 packed prefill against rectangular; equal-length cohorts lose. Source: [qualification/runs/sweep-20261006/lanes/history.md §1B](#)

A semantic activation capsule bus, and the false green it hid

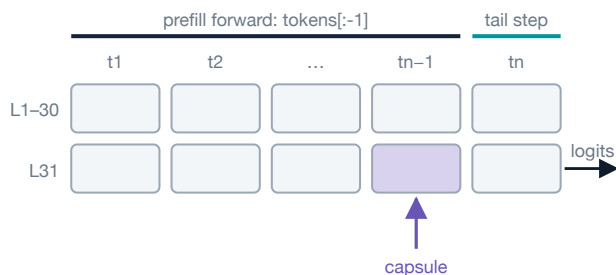
RESEARCH

UNQUALIFIED

NOT A PUBLIC HTTP API

The capsule bus (95caa1405) steers the model's activations instead of adding prompt tokens. Its M3 mechanism gate passed, but the sweep found that serving applied it at a different seam than the gate measured.

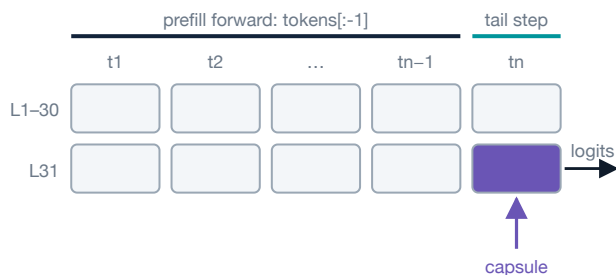
Before (CAP-2): capsule rode the prefill forward



nothing reads the last prefill row at layer 31

output bit-identical, yet the receipt said `observed_used=true`

After (cd330ca82): capsule rides the tail step



first logits change, as at the M3 gate seam

the receipt advances only from this evaluated forward

Where the capsule landed. Serving prefills all but the last prompt token, then feeds that token in a tail step that produces the first logits. A final-row intervention passed to prefill amended a row nothing reads at the last layer. The fix moves it onto the tail step with a one-step lifetime. Source: [qualification/runs/sweep-20261006/lanes/capsules.md](#) (CAP-2); cd330ca82

- Also fixed: a 0-error report on a waypoint 55% off (CAP-1), a one-shot capsule re-applied every step (CAP-3), a crash at step 0 (CAP-7), exact-prefix reuse for capsule requests (CAP-6), refusal at mount (ede895393).
- Verdict CAP-9: nothing in the server calls the bus yet; quality and performance qualification are open.

A laptop GPU runs on a power budget

The first equal-heat thermal map of mlx2: three routes loaded at once, alternated for 25 minutes from a cold start, with watts, P-states and die temperature on every segment.

89 w

cold burst, GPU

first 8K prefill: 892 tok/s at P12.5

40 w

sustained, GPU

prefill 620–686 tok/s for 25 minutes

63–67 °C

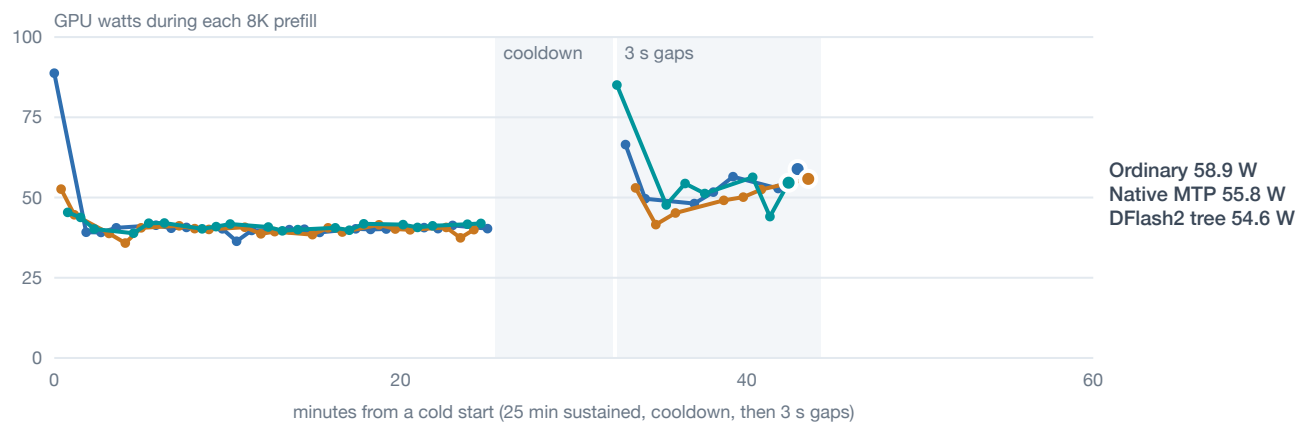
die plateau

below any thermal limit: a power budget, not heat

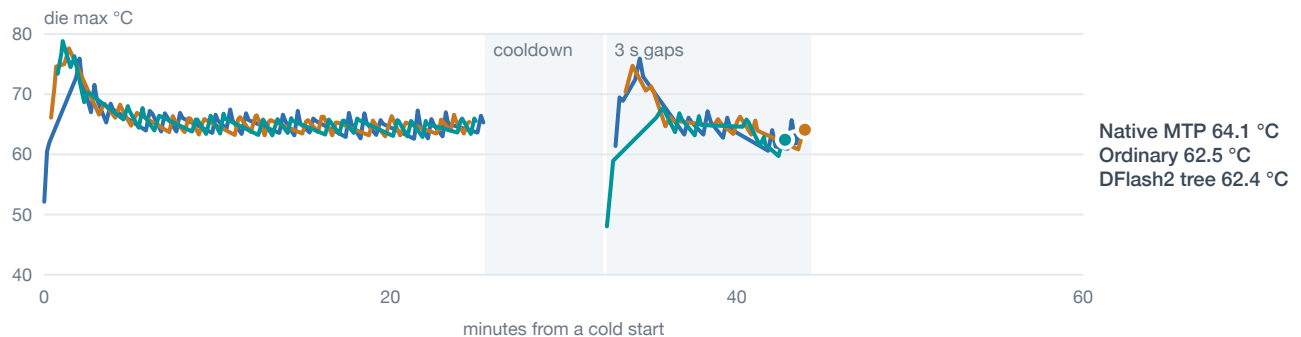
1.8–2.6×

B4 over B1, joules per token

batching is the efficiency lever



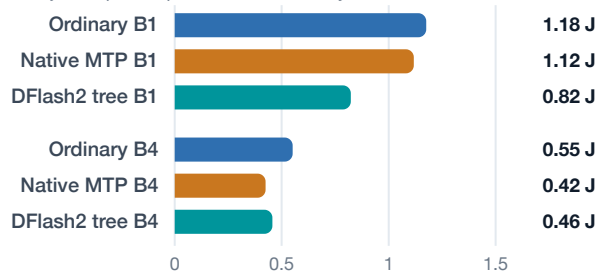
Burst, then budget. GPU power during each 8K prefill, three routes in strict alternation so every point is measured at the same heat as its neighbours. The first ~90 s run near 89 W; then the SoC holds about 40 W for the rest of the 25-minute hot phase. In the shaded phase each segment follows a 3 s idle gap. Source: [qualification/runs/sweep-20261006/thermal-q38/segments.jsonl](#)



Die temperature over the same window. It plateaus in the mid-60s while power is already capped, and recovers within minutes of rest. Source: [qualification/runs/sweep-20261006/thermal-q38/segments.jsonl](#)

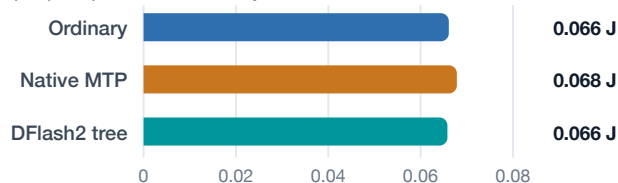
Decode, B1 then B4

GPU+DRAM joules per output token, hot steady



8K prefill

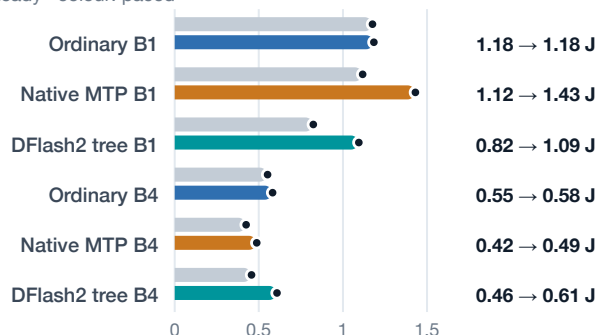
joules per prompt token, hot steady



What a token costs, hot. GPU + DRAM joules per token in the steady hot phase. DFlash2 is the cheapest B1 route, native MTP the cheapest at B4, and B4 costs roughly a half to a third of B1 on every route. Ordinary B1 decode is memory-bound: DVFS parks it near P7 on its own. Source: [qualification/runs/sweep-20261006/thermal-q38/summary.json](#)

Decode: steady vs 3 s gaps

grey: steady · colour: paced



8K prefill: steady vs 3 s gaps

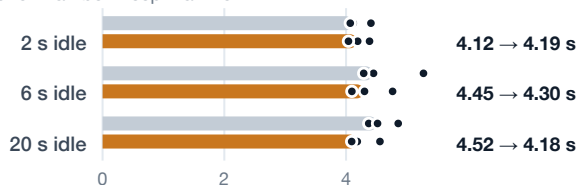
grey: steady · colour: paced



Pacing is not an efficiency lever. With 3 s gaps each segment runs at higher clocks and finishes a little faster, but a token costs +1 to +33% more energy (ordinary B1, already parked at low clocks, barely moves) and wall-clock throughput falls. The burst budget is a latency resource. Source: [qualification/runs/sweep-20261006/thermal-q38/summary.json](#)

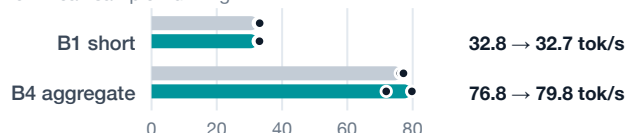
TTFT after an idle gap (4K prompt)

grey: ticker off · amber: keep-warm on



Decode with power telemetry on

grey: off · teal: sampler running

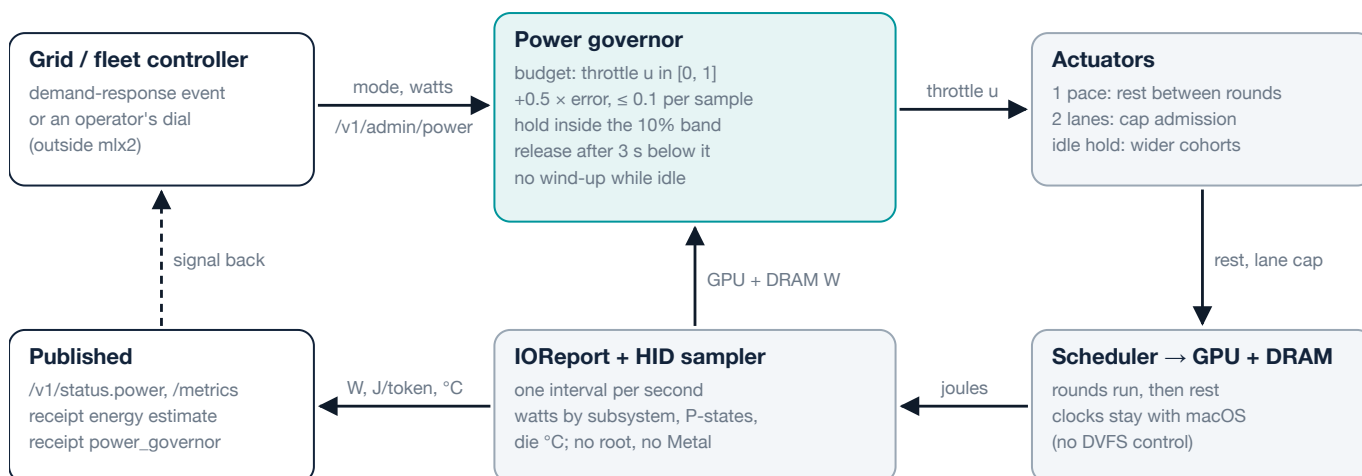


Two cheap questions, answered. Left: a keep-warm ticker saves about 0.3 s after 20 s idle on the M5 Max, not the 1.7 s a peer measured elsewhere, so it stays opt-in. Right: running the power sampler costs nothing measurable. Source: [qualification/runs/sweep-20261006/keepwarm/summary.json](#); [qualification/runs/sweep-20261006/telemetry-overhead/summary.json](#)

Publish the energy, then let the grid turn the dial

Telemetry says what a token costs; the governor holds a watt budget an operator or a grid signal can change at runtime.

If a server can say what each token costs in joules, and accept a ceiling at runtime, it can take part in grid demand response: shed load when the grid is stressed, run flat out when power is cheap and clean. mlx2 now does both halves. Power telemetry publishes the energy; the governor turns a requested watt budget into scheduling, without privilege and without touching clocks.

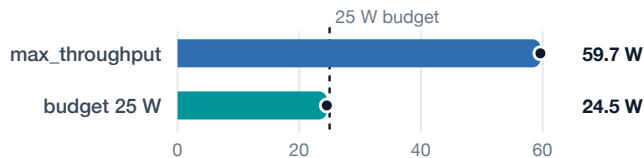


A dial for the grid. A controller outside mlx2 posts a mode and a watt budget to `/v1/admin/power`; the governor compares the sampler's GPU + DRAM watts with the budget and paces scheduler rounds first, then caps lanes. Everything it measures and does is published in status, metrics and each request's receipt. Source: docs/SERVING.md (power governor); 6dc3cfb03; 071fbb591

MODE	WHAT IT DOES	MEASURED ON QWEN3.8-27B
off	no governor object, metric family or receipt field	default
max_throughput	nothing actuated; the admin endpoint is live	59.7 W, 54.0 decode tok/s
efficient	idle hold up to 50 ms so near-simultaneous arrivals form wider cohorts	about -2 to -3% J/token on this workload governor
budget	closed-loop cap on rolling GPU + DRAM watts: pace first, then lanes	25 W budget held at 24.5 W

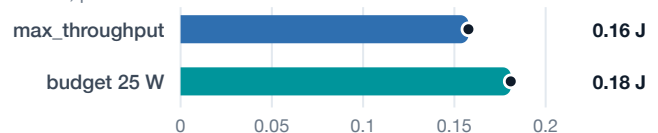
Average GPU + DRAM power

over the whole run



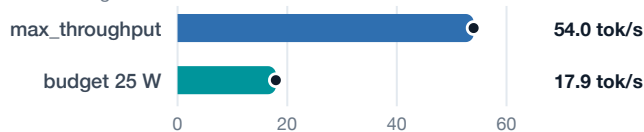
Energy per token

all tokens, prefill and decode



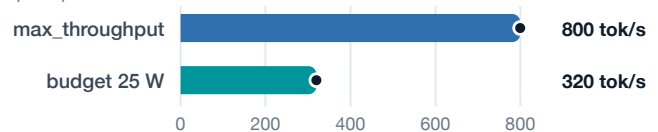
Decode rate

B1 and B4 segments

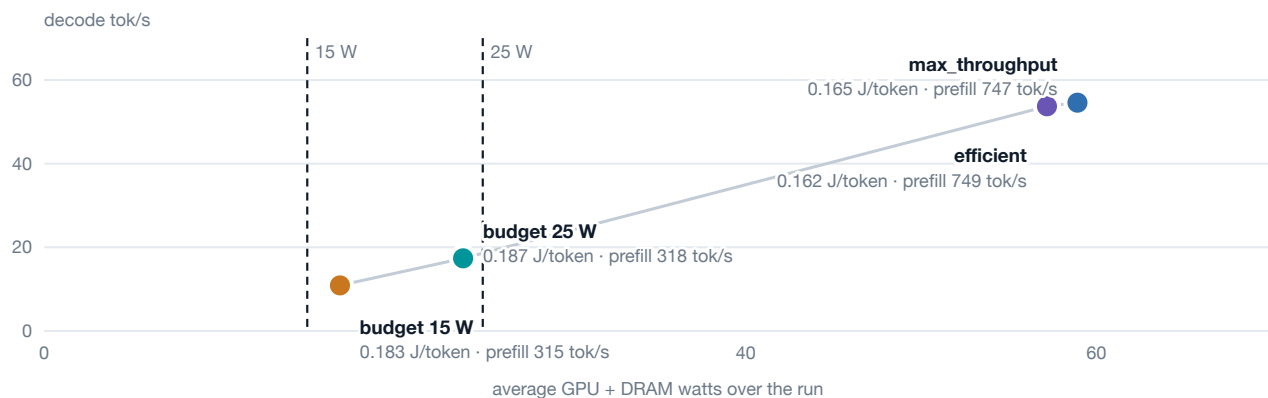


Prefill rate

8K prompts



A 25 W budget is a reliable hard cap. Final tree, strict alternation with max_throughput for 15 minutes, the cold first cycle excluded. Power fell 59%, decode 67%, prefill 60%, and each token cost 15% more energy: a demand-response lever, not an efficiency mode. Source: [qualification/runs/sweep-20261006/final-governor/segments.jsonl](#)



What each setting buys. Four governor settings on one tree (before the anti-windup fix), 20 minutes in strict rotation. The budget modes trade throughput almost linearly for watts; efficient sits next to max_throughput because this harness sends little co-arrival for it to merge. Source: [qualification/runs/sweep-20261006/governor/segments.jsonl](#)

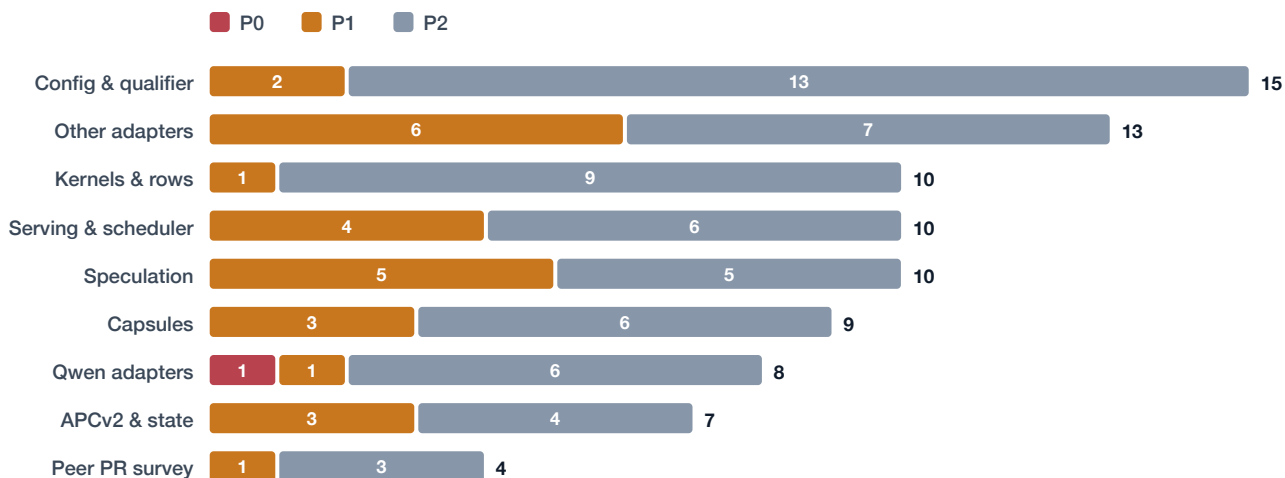
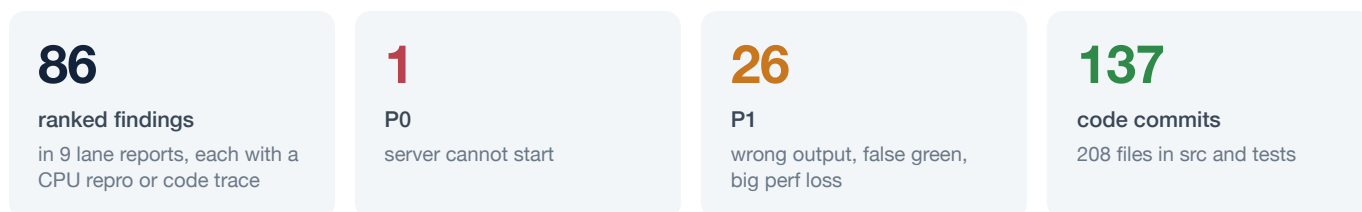
A CORRECTION WE MADE

The first run paced B1 decode to about 11 tok/s under 25 W, which we read as wind-up from host-wide power on an idle engine. We added anti-windup (f030c22fa) and re-ran. The real reason: B1 decode alone draws about 38 W of GPU + DRAM, so a 25 W budget legitimately throttles it. Anti-windup stays, because on a shared host idle wind-up is real. [FINDINGS-GPU.md](#)

- No clock control: macOS gives userspace none. The governor decides when the scheduler rests, how many lanes admission fills, and how long an idle server waits to form a cohort. Outputs change only through batch width, and receipts say so.
- Never starved: duty never drops below its minimum and one rest never exceeds 1 s; a stale power signal holds the throttle instead of releasing blindly.
- Power is whole-SoC: other work on the host counts against the budget. Default-off and unqualified.
- Next: a burst-reserve mode that keeps background load under the sustained level so interactive prefill can spend the 90 s burst; it needs request priority classes.

86 findings, 1 that stopped a server

A read-only audit by lane, repro-backed fixes in isolated worktrees, a review round, and the bugs that only showed up when everything met.



Findings per lane, by severity. Counted from report headings that carry an id and a severity; grouped ids and the seven paged-APC sub-findings count once. Source: [qualification/runs/sweep-20261006/lanes/*.md](#)

Nine code lanes read the tree at 187c6aa23 with no GPU and no edits, each writing failing CPU repros for its findings. Fix lanes worked in separate worktrees and merged into one integration branch; a review round then read the fixes, and the GPU campaigns ran on the baseline and again on the final tree. Six catches stand out.

P1 · CAP-2

The capsule that claimed to work

Serving applied the activation capsule on the last prefill row. At the shipped injection layer (31 of 32) nothing reads that row, so output was bit-identical to ordinary decode while the receipt said `observed_used=true`.

Found by tracing the serving seam against the M3 gate's seam and a tiny-model repro.

cd330ca82 — the capsule rides the prompt-tail forward; receipts advance only from that forward.

P0 · AQ-1

Qwen3.5-122B could not start

With the default `--lane-matmul auto`, the 122B loaded about 68 GB of weights and then died with an `AttributeError`: a lane hook added on 5 October read an attribute the 122B adapter never sets.

Found by an AST scan of the attributes each adapter class stores against the hooks serving calls.

f30cce87d, plus its own numerics contract and prefill default (5520c495a).

P1 · SS-2 / SS-3

A 9.5-second decode stall on Flash-Next

Beside a 16K prompt, a decoding lane on the served Flash-Next policy waited 6.9–9.5 s between tokens; ordinary waited 0.50–0.56 s. The stall bound kept a running maximum and self-MTP preparation was invisible to it.

Measured in the sweep's Flash-Next A/B; mechanism confirmed by CPU repros of the estimator.

d28a774aa + the co-arrival fixes: final max gap 440–475 ms, B8 median TTFT halved.

P1 · SPEC-10

Prompt lookup ran one lane at a time

On Qwen3.8's recurrent layers PLD could not batch, so four lanes ran four serial B1 rounds: 25–26 tok/s aggregate against 66–68 for ordinary, plus a replay forward on every partial acceptance.

Explained from the GPU route A/B down to the cache-topology check that refused hybrids.

2942ee30b, 82bdcc230 — batched, warm reuse fixed, B4 at parity with ordinary.

P1 · COW-1

The prefix cache paid twice

Every APCv2 entry published by a lane re-counted the recurrent snapshots it shares with its siblings: 3.25 MiB charged for 2.25 MiB held in the repro, up to about 5× on long hybrid prompts, so fewer entries fitted.

Found reading ArraysCache.nbytes against publication; failing CPU repro.

cc4e96bac — one physical-buffer ledger; evicting an entry frees exactly its charge.

P1 · CQ-4

Selected but never required

Several default-on mechanisms (Flash-Next fused GDN prefill and MoE weighted sum, lane matmul, base fairness) had counters but no engagement requirement, so a route could qualify without running them.

Found by diffing the default-on list against the qualifier's required checks.

734a79679 — required as run deltas; harness re-pinned.

Caught during integration

Five defects appeared only when the fixes met each other, main, or the GPU. Each was found by a failed start or a campaign number that did not make sense.

Integration · 9f9a81f86

Every guarded adapter refused to start

Parsing the new APC retention policy imported runtime.apc_v2, which loaded the model base before any adapter pinned its import-time environment. Qwen3.8-27B, Flash-Next and every other guarded adapter died with ImportError. A fresh-interpreter test now builds the adapter first.

Integration · e33356767

A main commit broke the default DFlash2 route

66b2f9d1e attached proposal composition to every external drafter, including the pinned tree pair, which refuses composition. The final route campaign could not start DFlash2. Composition now defaults on chain routes only; the A/B below shows the tree route back to its numbers.

Integration · b02360126

The snapshot ledger synchronised the device

The COW-1 fix keyed its ledger by evaluating arrays, forcing a GPU sync on every APCv2 store during prefill. The post-fix campaigns ran with it, which is why only the final tree's numbers are quoted as after.

Integration · e4213207f; 057373692

Co-arrived Flash-Next prompts admitted in waves

The first co-arrival hold bounded the wait by the siblings' sum, so eight prompts started in waves (TTFT 1.3 s, then 8.9 s, then 12.2 s). Bounded per sibling and prepared together, median B8 TTFT halved instead.

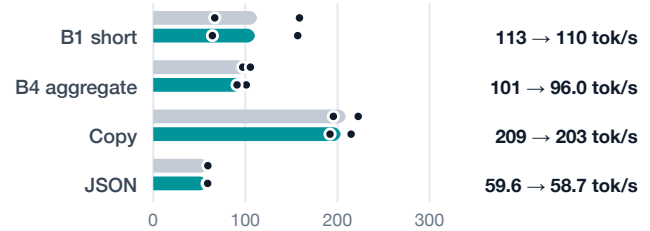
Integration · f030c22fa

The governor wound up on an idle engine

Whole-SoC power includes other servers' work; an idle budget server integrated it into its throttle. Anti-windup now holds the throttle while the engine is idle (see the correction in the governor chapter).

DFlash2 tree: composition off (grey) vs default

tok/s; dots are repetitions



The tree route after the composition fix. Default against an explicit `proposal_composition: false` on the pinned 27B tree: equivalent within repetition spread. Source: qualification/runs/sweep-20261006/dflash-comp/summary.json

The review-fix round

An independent code review of the integrated fixes raised findings that were fixed before the final campaigns:

AREA	FIX	COMMITTS
Prompt-lookup rollback	recurrent rows rewind on hybrid abort; PLD lanes rebuilt after a failed round	1d25dd7d1
Strict acceptance compare	block and self-MTP verification compare uniforms strictly	8ca62ebf5
Proposal laws validated first	full validation before the greedy verify shortcut	91073ecad
Native sampler safety	IOReport/HID results checked, owned objects released, sampling serialised	f52578e61; f03a51b6e
Governor signal integrity	only complete GPU + DRAM readings feed the controller; window bounded by seconds	dffad06da; 01fd7a3a2
APCv2 accounting	only freeable bytes counted; a refused hit's depth and reuse count withdrawn exactly	b310c3e13; 73548bdba
Stall-bound hygiene	stale cost-model samples expire; one tile probed when there is no measurement	8c7e1fe78; 7502a23ca
MoE pad counters	every pad counter resets on <code>moe_pad_status(reset=True)</code>	6cf03583e

READINGS WE WITHDREW

- **A route prefill penalty** on the 27B was heat: the first ordinary run was cold (8.4 s), every hot run took about 10 s whatever the route.
- **A Flash-Next B8 "regression"** (143–159 → 87 tok/s) was a metric artifact: first-token-to-last aggregate rewards staggered starts. Over batch wall time throughput is unchanged and TTFT halved.
- **Governor wind-up** on B1 decode was the budget doing its job (chapter 4).

FINDINGS-GPU.md

The external review, answered

An external technology review of the same source (6 October) ranked ten next steps. The sweep's response to each (sources: MLX2 Technology & Research Review, 6 Oct 2026, §1; qualification/runs/research-20261006*/README.md; commits named):

#	REVIEW PROPOSAL	WHAT THE SWEEP DID	STATUS
1	Stabilize the numerical route before restoring faster packed prefill	The pinned Qwen3.8 default keeps external varlen prefill and dense-MLP compaction off; the pair is now one operator-owned default group (29056b2fb). Not restored.	HELD
2	Dispatch MoE tiles from the actual routing distribution	Measured: routing CV 1.87, yet the planner is within 3–5% per call of the best plan; about 0.5% of a 35B 8K prefill is recoverable, under the 3% bar. Not built.	CLOSED NEGATIVE
3	Completion-timed, geometry-aware prefill bounds	Shipped as the default stall estimator (d28a774aa). Same-code GPU A/B: neighbour p95 –16 to –23%, worst gap –13 to –18%, for +1 to +6% long-prompt TTFT.	NEW DEFAULT
4	Exact self-sampling QSA top-k (GVR)	Merged default-off (a768539bf), exact. Synthetic: selector 0.75×, stage one 0.83× of direct8; heavy ties fall back slower. Not selected.	MERGED, NOT SELECTED
5	Exact stochastic block verification	Opt-in on the linear external chain (ccbf2a99c) with an exhaustive exactness oracle; GPU A/B at T = 0.7 pending.	OPT-IN
6	Retain APC state by saved compute per unique byte	Opt-in value-aware retention (f44f355b5) on the new shared-buffer ledger; CPU replay only.	OPT-IN
7	Correct capsule reconstruction objectives	CAP-1 fixed: retained waypoints stay exact and the reported error measures discarded movement (e5514a5d0).	FIXED
8	Narrowly scoped routed A8 gate+up	Amdahl bound: routed gate+up is 24.8% of a 35B 8K prefill; A8 at 1.3–1.5× gives +6.1–9.0% on all layers or +2.4–3.5% on NAX 4-bit layers only. Approximate; not built.	ESTIMATED
9	Conditioned block algebra for GDN	Chunk-16 prefill is 1.19–1.46× per layer at 27B shapes but inexact (state error 4e-3, 0.8–1.6% argmax flips) for about a 1% prize. Not selected.	CLOSED NEGATIVE
10	Optimize KV precision for output error and total bytes	Not addressed in this sweep.	OPEN

What the sweep added, not just repaired

Several findings were best answered with a mechanism. Each is listed with whether it is on, behind a flag, and what the GPU said about it.

Completion-time stall estimator

NEW DEFAULT

REQUAL ADVISED

Prefill slices beside live decode are sized by $t = a + b \cdot R$ per route kind and KV depth; self-MTP preparation is timed and owes debt (SS-2/3/8/10). Same-code GPU A/B: neighbour p95 -16 to -23%, worst gap -13 to -18%, long TTFT +1 to +6%.

d28a774aa; acadd8b76; estimator campaign

Stall bounds on the external and PLD loops

ON

The external-draft and prompt-lookup routes ran their own prefill loops with fairness off. They now apply the same bound: DFlash2 neighbour p95 693-767 → 463-469 ms, PLD 703-710 → 492-508 ms.

673aa9fe0; 7502a23ca; d0ac480ae

Co-arrival hold for self-MTP cohorts

BEHAVIOUR CHANGE

Multi-chunk prompts that arrive together form one MTP cohort (SS-1), bounded per sibling; partial first boundaries are prepared together. 27B MTP B4: 46-51 → 104-111 tok/s.

56fce148e; e4213207f; 057373692

6-bit affine NAX sorted gather

ON BY DEFAULT

REQUAL ADVISED

Admits 6-bit experts to the fused MoE gather (widths 4/6/8). Bit-identical logits; Qwen3.6-35B 8K TTFT -7%, B8 +3 to +7%.

MLX2_MOE_NAX_GATHER_BITS=4,8 restores the old set.

7e28b1ace; 46656a8ec; final-q36-6bit

Power and thermal telemetry

OPT-IN

OVERHEAD UNMEASURABLE

--power-telemetry publishes watts, joules, P-state residency, die temperatures and a rolling J/token without root. Same-code A/B: B1 32.8 vs 32.7 tok/s, B4 within noise.

a809a970d; f52578e61; telemetry-overhead campaign

Power governor with a grid dial

DEFAULT-OFF

UNQUALIFIED

max_throughput, efficient and budget modes; a runtime admin endpoint sets mode and watts. A 25 W budget held 24.5 W on the GPU. Anti-windup while idle.

6dc3cfb03; 071fbb591; f030c22fa

Prompt lookup batched on hybrid targets

OPT-IN ROUTE B4 AT PARITY

One verify forward per round on GDN hybrids, trim instead of replay, the cohort kept whole (SPEC-10). B4 rose only to parity with ordinary (55-63 tok/s) once plain rounds were pipelined.

2942ee30b; 1c45a38a7; 82bdcc230

PLD warm prefix reuse

OPT-IN ROUTE MEASURED

Prompt-lookup prefill records recurrent checkpoints, so a shared document is a warm hit: 8K warm TTFT 9.7 → 1.6 s with 6,144 tokens cached.

cb26c206b; final-q38-routes

Sidecar-ancestor hits

ON

Draft routes walk to the deepest resident ancestor that carries a usable draft sidecar (APC-1); a spilled draft checkpoint restores as that ancestor.

f6a160933; 1ee34dd68

Shared snapshots charged once

ON

A physical-buffer ledger stops double-charging aliased recurrent snapshots (COW-1), keyed without evaluating arrays after the first version forced a device sync per store.

cc4e96bac; b02360126

Value-aware APCv2 retention

OPT-IN

Evict least saved prefill per freed unique byte first; withdrawn hits give their credit back.

f44f355b5; cd9fb13b3

Exact block verification

OPT-IN

Block verification for the linear external chain with an exhaustive RNG-branch oracle and strict uniform compare.

ccb2a99c; 883c0b638; 8ca62ebf5

GPU keep-warm ticker

OPT-IN, STAYS OFF

--gpu-keep-warm-seconds N ticks a one-element kernel after real work. Measured wake penalty on M5 Max: about 0.3 s after 20 s idle, not the 1.7 s a peer reported.

726837c12; keepwarm campaign

GVR-style exact QSA selector

DEFAULT-OFF NOT SELECTED

Self-sampling multi-threshold top-k, exact against the oracle. Synthetic: selector 0.75×, stage one 0.83× of direct8; heavy ties fall back slower.

41b87aa11; 223068567; provenance/qsastage1-gvr-selector-2026-10-06.json

Claude Code tool search

AGENT COMPAT

defer_loading tools and tool_addition / tool_removal / tool_definition blocks resolve into the tool list instead of a 400.

fc7ca8777

Qualifier requires default-on engagement

HARNESS RE-PINNED

Flash-Next fused GDN prefill and MoE weighted sum, covered lane matmul and base fairness must now run to qualify (CQ-4); behaviour-changing MLX2_* switches enter settings and the APCv2 key (CQ-13).

734a79679; e385f6be5

DEFAULTS THIS SWEEP CHANGED

The stall estimator, the external-route stall bounds, the self-MTP co-arrival hold, 6-bit NAX gather, warm-continuation slicing, sidecar-ancestor lookup and shared-snapshot charging change behaviour on default routes; every opt-in above leaves defaults untouched. Routes touched by a default change need requalification before they carry the qualified label again.

What the final campaigns did not fix

Measured limits, by-design behaviour and the qualification work the new defaults require.

OPEN ITEM	WHAT WE KNOW	EVIDENCE	STATUS
Prompt lookup B1 overhead	B4 now matches ordinary (55–63 against 62–63 tok/s on the closing run, up from 25–33), but single-stream PLD still runs about 10% below ordinary when it rarely proposes.	FINDINGS-GPU.md (final-pld)	OPEN
Native-MTP warm hit depth is unstable	One repetition resumed from 4,096 cached tokens (TTFT 4.2 s), the other from 7,168 (0.64 s). Flash-Next ordinary once hit only 8,192 of a 16K prefix.	final-q38-routes; final-fn	OPEN
Worst-case gaps still exceed the target	p95 meets about 500 ms on every route, but single waits reach 613–1,010 ms on native MTP, 533–815 ms on prompt lookup and 509–636 ms on DFlash2.	final-q38-routes	OPEN
DFlash2 warm reuse	Warm 8K TTFT stays near 8.6 s: the external route pairs draft state only at the prompt boundary, so a shared prefix cannot be served from the cache.	final-q38-routes; FINDINGS-GPU.md	BY DESIGN
Composition default on chain routes	After the fix, proposal composition defaults only on chain routes (Muse DFlash2, North EAGLE). Its effect there is unmeasured.	dflash-comp; e33356767	NEEDS QUALIFICATION
Speculative output identity	Speculative outputs differ textually from ordinary on some prompts; they need near-tie classification, not a token compare.	q38-routes; docs/QUALIFICATION.md	CLASSIFY
M3-only test failures	The full suite passes on the M5 at the final tree (11,646 tests, 42/42 import guards). The M3 fails only tests that need mlx_vlm, native binaries or the TensorFold checkout.	FINDINGS-GPU.md	ENVIRONMENTAL
Requalification of changed defaults	The estimator is now part of route identity; the NAX 6-bit default, the external routes' fairness and the repinned qualifier harness all change what a receipt certifies.	acadd8b76; 7e28b1ace; 673aa9fe0; 734a79679	REQUAL NEEDED
Burst reserve and governor qualification	The governor is default-off and unqualified. A burst_reserve mode (keep background load under the sustained level so interactive prefill can spend the burst) needs request priority classes the scheduler does not have.	docs/SERVING.md (power governor)	NOT BUILT

WHAT "AFTER" DOES NOT MEAN

The final numbers come from two repetitions on one laptop at controlled but not identical heat. They show direction and size, and they are enough to choose defaults; quoted performance claims need a separate controlled run.

WHAT IS NOT ON THIS LIST

Anything that measured within noise and was rejected (GDN chunk-16 prefill, top-8 routed MoE decode, a histogram MoE planner, LLaDA) is closed, with its evidence kept, rather than open.

Everything else the week shipped, with its status

Defaults, opt-ins, research routes and an honest closure. Numbers are as recorded by the agents that ran them; the history digest names each run.

QSA direct4 / direct8 selectors

QUALIFIED EXACT UNSELECTED

Exact selector alternatives remove about 1 GiB of selector scratch at 8192 × 32768. Loaded-model throughput moved +0.86% (direct4) and −2.93% (direct8), so radix_exact stays selected; the sweep's Flash-Next A/B found direct4 neutral.

714af9653; qualification/runs/sweep-20261006/lanes/history.md §1B; fn campaign

Prompt-length prefill autoscale

ON BY DEFAULT

An unset prefill step scales with the prompt: up to 32K 512 rows, up to 64K 2048, above that 8192. Operator beats adapter beats autoscale. The 8192 rung is unqualified at 128K (a 128K/4096 run swapped twice).

d31971dda; qualification/runs/sweep-20261006/lanes/history.md §1B

Native paged KV, N1–N20

RESEARCH ROUTE OBSERVED-USED

A real varlen/paged attention route with private KV and GDN owners. A B20 domain cohort completed 20/20 (139,531 prompt tokens) but its ordinary control had admission timeouts, so there is no valid comparison; at deterministic arbitrary traffic it was 7.44% slower overall.

docs/research/2026-10-04-agent-handover.md; qualification/runs/sweep-20261006/lanes/history.md §1B

Hybrid packed prefill

RESEARCH

Packed prefill for hybrid GDN/attention stacks. Stage profile: MLP is 57.9% of prefill and fences cost about 3.2 s; MLP one-eval, BF16 dequant, tiled SwiGLU and packed gate/up were each measured and rejected.

docs/research/2026-10-04-hybrid-packed-prefill-*.md; qualification/runs/sweep-20261006/lanes/history.md §1B

RaggedVerifyLayout contract

CONTRACT ON, SELF-MTP

One authoritative ragged-verify geometry and an exact row-capability contract, after vLLM/SGLang lessons and oMLX #4105. Scoped qualification for the Qwen3.8 B4 self-MTP physical varlen gate; no default speed change.

runtime/ragged_verify_layout.py; qualification/runs/sweep-20261006/lanes/history.md §1B

Qwen4 sparse-MoE row compaction

DEFAULT-OFF

Removes padded rows from Flash-Next's routed experts. A 4 × 32K 16-lane TensorFold + QSA composition was not memory-safe (swap-outs), so the composition gate is open.

79e2eae92; provenance/varlen-sparse-moe-qwen4-2026-10-04.json

Exact-prefix strategy transfers

CONTRACTS ONLY

Longest-first and prefix-cascade proposal strategies, transferred to twelve families as contracts and gates. On Qwen3.8: 1.33× at 2K, 1.085× at 8K, 1.062× at 32K, 1.001× at 64K; a retention cycle was fixed (38.5 → 18.8 GiB at 16K).

qualification/runs/32k-proposal-strategy-scout-73bf6c43d-20261005; qualification/runs/sweep-20261006/lanes/history.md §1B

TensorFold 0.6.1 long-context intake

ON BY DEFAULT **ROUTE REQUAL PENDING**

Fused QSA block scores, early PLE dispatch and row-exact batched fused GDN verify are on by default for Flash-Next. A matching qualified receipt at the current profile does not exist yet.

provenance/tensorfold-0.6.1-flashnext-longctx.json; qualification/runs/sweep-20261006/lanes/history.md §1A

Flash-Next roadmap ports

ON BY DEFAULT QUALIFIED ON E963AAED

Batched fused GDN decode, routed MoE decode, attention rows, HC decode kernels with 8-bit support and launch-time top-k fold, ported from oMLX. Row-exact verify stayed off: its copy windows ran at 0.78× of stock MTP.

qualification/runs/sweep-20261006/lanes/history.md §1A;
qualification/runs/omlx-combined-rowexact-20261001

Approved default flips (2 Oct)

ON BY DEFAULT

Adaptive MoE pad, decode-first order, copy span 14, QSA NAX crossover 8192 and per-model handoff widths for Flash-Next, 27B and Qwen3.6. The 27B all-features ladder still fails at 65K width 4 (memory admission).

e963aaed; qualification/runs/sweep-20261006/lanes/history.md §1A

Mechanism intake (92 commits)

DEFAULT-OFF PROBES

Peer-derived probes: external-draft allocator reclamation (4.30 GB pool → 8 MB), prefill reclamation (~81% pool), qSDPA query-row tiling at 154K (1.31 → 0.43 GB). Top-8 routed MoE decode was 12% slower and stays unselected.

qualification/runs/mechanism-intake-20260930/summary.json

Compiled PLE chains

ON BY DEFAULT

Flash-Next's learned n-gram embedding chains compile on a repeated signature and evict least-recently-used entries instead of overflowing the compile cache.

d7ee03da3

mlx2-top live metrics

TOOL

The monitor shows live inference metrics in place of the old accelerator panel; this sweep added unprivileged IOReport power, P-states and die temperatures as its default source.

28e615f0e; b2367195d

Near-tie qualification rule

RULE ON

Same top two, both margins ≤ 0.5 nat, shared prefix, functional oracle passing: classified a near tie, not a failure. B1 and state gates stay strict. The sweep tightened it after SPEC-02 showed it passed real divergences.

docs/QUALIFICATION.md; qualification/runs/sweep-20261006/lanes/history.md §1B

LLaDA direct denoising lane

CLOSED NEGATIVE

Active-block postprocess and active-head projection reached exact component parity but no worthwhile gain (slower at 4K, thermally invalid). Closed unless new evidence; the sweep refuses prompts containing the mask id.

9f06f85b8; 330104147; 71b22db4f

Parallel drafting (XPress, LiLiCorr, DPara)

RESEARCH DEFAULT-OFF

Many proposers under exact verification, including a 15-continuation pool. A real Qwen3-4B + XPress M3 smoke passed; no speedup was demonstrated and the pool adds large overhead.

docs/research/parallel-draft-candidates.md

Incremental tokenizer v1

OFF

Template plus encode 24× faster on 100K-character prompts, with no GPU decode benefit yet; production use needs an identity and invalidation design.

qualification/runs/sweep-20261006/lanes/history.md §1B

Host memory floor for admission

OPT-IN SWEEP FIX

An opt-in floor of free host memory that lane admission keeps (2 Oct). The sweep found that the $n > 1$ parallel-sample guard read headroom without the floor; both now share one policy-shaped headroom function.

c842d6210; fc22e2885;
tests/test_sweep_20261006_parallel_sample_floor.py

Every campaign, and the baseline runs behind the story

The campaign register, then baseline results the earlier chapters only cite.

The feature profiler drives a real server per arm through the HTTP API, collecting receipts, counter deltas, latency, swap, energy per token (IOReport GPU + DRAM) and output identity. The thermal map loads every arm at once and alternates them strictly. This register lists every campaign the book reads.

CAMPAIGN	SET	MODEL	ARMS	SHAPE	SWAP-OUTS
q38-routes	before	Qwen3.8-27B-oQ4e-mtp	ordinary, mtp, dflash, pld	2 reps	0
fn	before	Qwen3.8-Flash-Next-Uncensored-MLX2-4bit-MTP	served, ordinary, served_qsa_direct4	2 reps	0
q38-features	before	Qwen3.8-27B-oQ4e-mtp	served, lm_off, no_interior, int8_mlp, gdn_chunk16	2 reps	0
q36	before	Qwen3.6-35B-A3B-Heretic	mtp, ordinary	2 reps	0
thermal-q38	before	Qwen3.8-27B-oQ4e-mtp	ordinary, mtp, dflash	thermal map	–
estimator	intermediate	Qwen3.8-27B-oQ4e-mtp	mtp_cost, mtp_runmax, ord_cost, ord_runmax	2 reps	0
keepwarm	intermediate	Qwen3.8-27B-oQ4e-mtp	off, keepwarm	3 reps	0
telemetry-overhead	intermediate	Qwen3.8-27B-oQ4e-mtp	off, telemetry	3 reps	0
governor	intermediate	Qwen3.8-27B-oQ4e-mtp	maxtp, efficient, budget25, budget15	thermal map	–
post-q38-routes	intermediate	Qwen3.8-27B-oQ4e-mtp	ordinary, mtp, dflash, pld	2 reps	0
post-fn	intermediate	Qwen3.8-Flash-Next-Uncensored-MLX2-4bit-MTP	served, ordinary	2 reps	0
final-q38-routes	final	Qwen3.8-27B-oQ4e-mtp	ordinary, mtp, dflash, pld	2 reps	0
final-fn	final	Qwen3.8-Flash-Next-Uncensored-MLX2-4bit-MTP	served, ordinary	2 reps	0
final-q36-6bit	final	Qwen3.6-35B-A3B-Heretic	nax48, nax468	3 reps	0
final-governor	final	Qwen3.8-27B-oQ4e-mtp	maxtp, budget25	thermal map	–
dflash-comp	final	Qwen3.8-27B-oQ4e-mtp	dflash_default, dflash_nocomp	2 reps	0

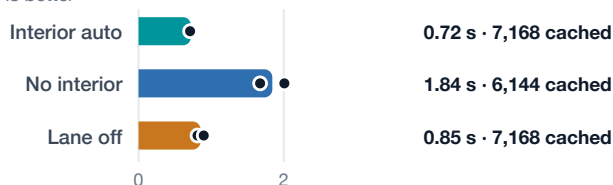
Qwen3.8-27B, every route metric, baseline → final

ROUTE	B4 AGG TOK/S	GAP P95 MS	GAP MAX MS	WARM 8K TTFT S	B1 TOK/S	COPY TOK/S
Ordinary	66–68 → 70–75	548–651 → 424– 451	714–728 → 449–482	1.89–1.91 → 1.57– 1.63	32–33 → 33	31–32 → 32
Native MTP	46–51 → 104– 111	572–612 → 446– 470	1033–1074 → 613– 1007	0.55–0.82 → 0.64– 4.23	48–63 → 53–64	135–147 → 146– 148
DFlash2 tree	97–101 → 103– 105	759–789 → 463– 469	805–851 → 509–636	9.86–10.26 → 8.60– 8.65	56–135 → 66– 158	183–206 → 195– 217
Prompt lookup	25–26 → 26–33	739–746 → 492– 508	765–824 → 533–815	9.54–9.87 → 1.55– 1.60	26–27 → 29–30	143–145 → 157

Ranges over repetitions. Source: qualification/runs/sweep-20261006/q38-routes/results.json; qualification/runs/sweep-20261006/final-q38-routes/results.json

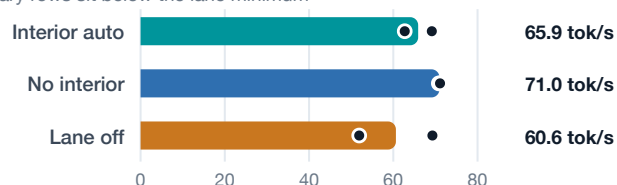
Warm 8K prefix: time to first token

lower is better



B4 aggregate decode

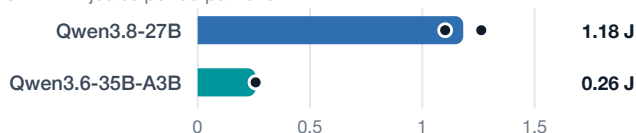
ordinary rows sit below the lane minimum



Qwen3.8-27B ordinary route: interior checkpoints and lane matmul (baseline). Interior checkpoints let the warm request resume deeper in the shared document. Lane matmul changes nothing on ordinary B1/B4, whose row counts sit below its minimum; it matters for verify rows. A follow-up found int8 prefill refused by this adapter (correctly) and GDN chunk-16 within noise and inexact, so both stay off. Source: qualification/runs/sweep-20261006/q38-features/summary.json; FINDINGS-GPU.md (q38-features2)

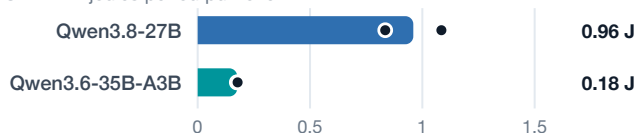
Short chat, B1

GPU+DRAM joules per output token



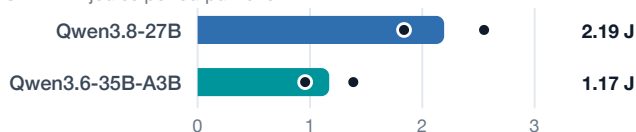
Batch (27B ×4, 35B ×8)

GPU+DRAM joules per output token



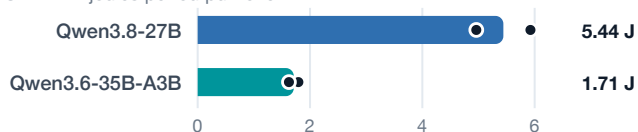
Warm 8K prefix

GPU+DRAM joules per output token

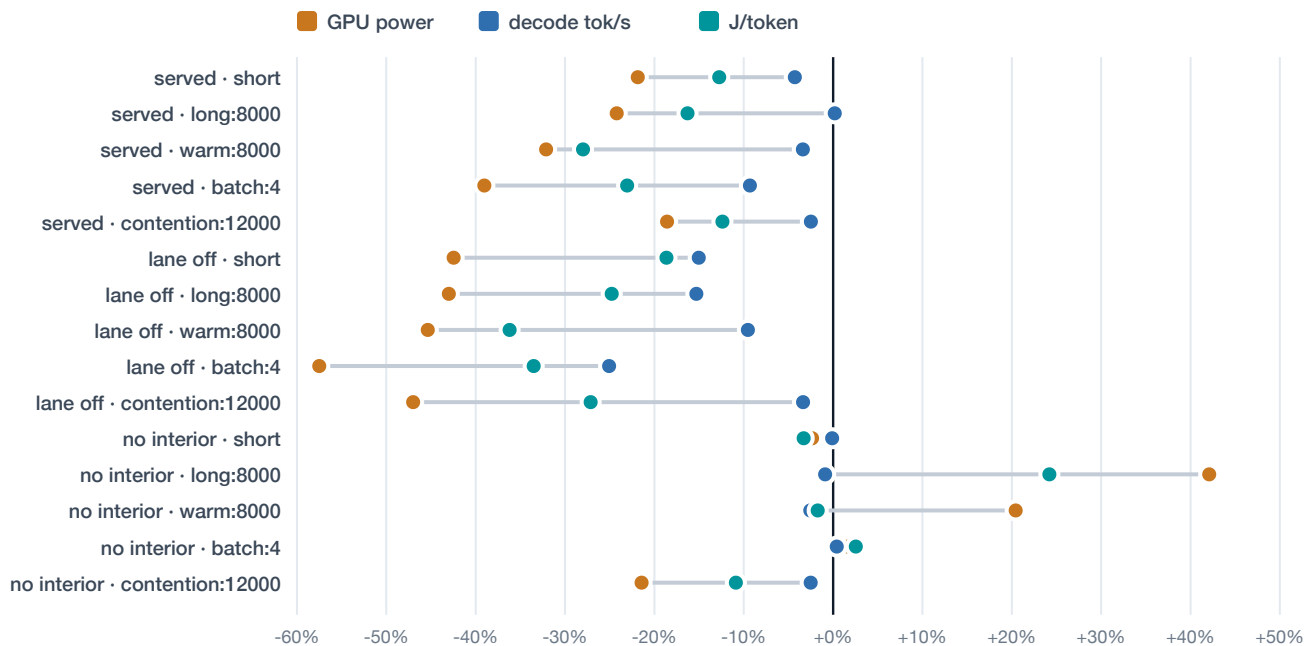


Long prompt (8K / 16K)

GPU+DRAM joules per output token



Energy per output token on ordinary routes (baseline). The 35B-A3B mixture of experts activates about 3B parameters per token and spends a fraction of the dense 27B's energy. Long-prompt rows include prefill energy divided by a 128-token answer. Source: qualification/runs/sweep-20261006/q38-features/results.json; qualification/runs/sweep-20261006/q36/results.json



Heat inside an A/B campaign. Change from the first to the second repetition of the same arm on the baseline 27B features run: later, hotter repetitions drew much less GPU power for a small loss of throughput, so energy per token improved. Clock policy moved these numbers; that is why the thermal map and governor runs alternate arms at equal heat. Source: [qualification/runs/sweep-20261006/q38-features/results.json](#)

DEPLOYMENT DRIFT

The live 27B service had been running a compiled-replay policy from an unmerged worktree; main's adapter refuses that policy. The service is stopped; its policy is not part of any number in this book. [FINDINGS-GPU.md](#)

Earlier evidence still standing

400 of 400 domain checks, M5

Four-lane 20×20 batching on Qwen3.8 MLX 4-bit passed with APCv2 repeated-prefix reuse, zero HTTP errors and zero swap-outs; median aggregate generation 46.8 tok/s across the mixed workload.

[benchmark_results/2026-09-29/qwen38-mlx-4bit.md](#) · September source

Cross-host stress pass, M3

The same 20×20 passed 400 of 400 on an M3 Pro with APCv2 reuse and zero swap-outs; median aggregate 13.8 tok/s.

[qualification/runs/requal-20260928/README.md](#)

32K warm reuse depends on capacity

An 8 GiB APCv2 cap missed one of four warm lanes in each of three width-four repetitions; a 16 GiB cap passed all four lanes, three repetitions and 30 needles.

[benchmark_results/2026-09-29/qwen38.md](#)

Offline diarization, qualified

Nemotron 3 speaker turns over 248 recordings (52.6 h): DER 0.0844 vs 0.0828 reference on VoxConverse and 0.1136 vs 0.1137 on AML.

[docs/NEMOTRON3-DIARIZATION-QUALIFICATION.md](#)

The runtime, feature by feature

Each capability connects its purpose to its implementation and operating boundary.

The full capability map: 129 capabilities, 18 new this week **NEW** and 9 updated **UPDATED**. Inclusion means implemented behaviour or an explicitly identified candidate; it does not label the inventory qualified. The screen edition expands each card's mechanism and boundaries.

Product surface API · clients · outputs · 27

AGENTS **NEW**

Claude Code tool-search blocks

Deferred tools and mid-conversation tool_addition / tool_removal / tool_definition blocks resolve into the offered tool list. *Claude Code with tool search stops getting a 400 and silently retrying without its tools.*

[translation](#)

SERVING

OpenAI-compatible chat and text completions

Chat/completions and completions share one bounded request lifecycle, with nonstreaming and SSE streaming responses.

Existing clients can move onto Apple Silicon without giving up a familiar API shape.

[serving guide](#) [server](#)

STREAMING

SSE, cancellation and backpressure

Disconnects and slow consumers cancel work; output queues, connections and inflight requests are bounded. *Runaway clients do not silently become runaway GPU work or unbounded memory.*

[lifecycle](#) [implementation](#)

AGENTS

Function calls and Qwen tool parsing

OpenAI-shaped tools, auto / none / required / named choice and adapter-owned parsers integrate with agent loops. *Local inference can participate in real tool-using workflows rather than remain a text-only demo.*

[parser](#) [contract](#)

REASONING

Hermes and thinking controls

options, think, enable_thinking, template controls and effort labels normalize into explicit route controls; conflicts and unknowns fail. *Different clients get one deterministic control language instead of accidental template behavior.*

[controls](#) [adapter](#)

SAMPLING

Full per-request sampling controls

Temperature, top-p, top-k, min-p, seed, stop strings, token caps, logit bias and repetition/presence/frequency penalties flow through ordinary and speculative target laws. *Optimized routes preserve the semantics applications expect from the reference route.*

[sampling](#) [validation](#)

PROBABILITY

Actual emitted-token log probabilities

Logprobs and up to eleven ranked alternatives are serialized for ordinary, accepted-draft and correction/bonus tokens in both response modes. *Scoring, confidence and audit consumers see the target distribution that produced the output—not the draft distribution.*

[semantics](#) [implementation](#)

CONSTRAINTS

Regex, JSON object and strict JSON Schema

Bounded token automata and an exact scanner fallback enforce supported output languages, with explicit error and qualification boundaries. *Structured generation has a bounded, inspectable enforcement path instead of relying on prompt wording.*

[engine](#) [contract](#)

PARALLEL OUTPUT

n=1...8 atomic sampling cohorts

The whole cohort is reserved before publication; independent seeds branch from one attested APCv2 prompt boundary.

Multiple choices pay for one prompt when safe, while partial allocation and misleading “shared prefill” claims are rejected.

[tests](#) [fanout tests](#)

RESPONSES

OpenAI Responses API with bounded state

/v1/responses runs on the chat lifecycle with typed SSE events, function tools, a tenant-scoped store, previous_response_id continuation and cursor-shaped input items. *Clients written for OpenAI's newer API, including stateful multi-turn loops, run locally without a second engine.*

[translation](#) [parity](#)

ANTHROPIC

Anthropic Messages endpoint

`/v1/messages` and `/v1/messages/count_tokens` translate system, text, tool_use/tool_result and thinking blocks onto the shared engine, with Anthropic-shaped errors and SSE. *Anthropic-SDK clients and agent harnesses get the same admission, receipts and tool contract as OpenAI clients.*
[translation](#) [contract](#)

REASONING

HMAC-signed reasoning replay

Returned thinking blocks and Responses reasoning items carry an HMAC bound to model and tenant, and only verified reasoning re-enters the prompt on a later turn. *Stateless clients can hand reasoning back without letting a forged or cross-tenant trace steer the model.*
[signer](#) [semantics](#)

CLIENTS

Official OpenAI and Anthropic SDK smoke

`scripts/sdk_smoke.py` drives the real openai and anthropic Python packages against a scripted CPU server or a live server, gated on the capabilities `/v1/status` declares. *Compatibility is checked with the client libraries people actually use, not only hand-written JSON.*
[smoke](#) [coverage](#)

LIMITS

2M-token caps with prompt-aware default

Explicit output caps accept up to 2,097,152 tokens and are never clamped; an omitted cap resolves to the smaller of `--default-max-tokens` and the remaining context. *Long generations need no guessed cap, and operators can trade the default length for admission width.*
[limits](#) [admission](#) [projection](#)

SAMPLING

Vendor sampling defaults per model

Each adapter declares its vendor's recommended sampling profiles, cited to `generation_config.json` and the model card; unset request fields take the profile for the resolved thinking mode. *Models run with their authors' recommended sampling by default, and the receipt shows where each value came from.*
[profiles](#) [vendor defaults table](#)

REASONING

Thinking budgets and run-on release

A per-request or effort-scaled reasoning budget, a content-blind run-on alarm and a ramped bias on a verified close-token sequence let a model finish thinking instead of being cut off. *Reasoning models can finish a formed answer instead of being cut off mid-thought. The mechanism ships and is unit-tested; it currently has no valid quality measurement.*
[guard](#) [what the norm fix voids](#)

STEERING

Artifact-calibrated reasoning steering

A direction calibrated on the served artifact can steer supported reasoning lanes toward their close, under independent evidence and cache-fidelity rules. *Reasoning control is tied to measured artifact behavior rather than a transferable magic vector.*
[calibration](#) [re-measurement](#)

CONSTRAINTS

Grammar deferral past thinking

A structured request can think first: the grammar starts after the adapter's thinking-close marker, leaving reasoning free while enforcing the supported answer format. *Reasoning models no longer have to choose between thinking and schema-valid output.*
[tests](#) [structured output with thinking](#)

AGENTS

Forced tool calls constrained by grammar

Under the `constrained_tool_grammar` policy, required and named tool choices decode under the adapter's own call-format grammar instead of only being checked afterward. *Decode-time constraints reduce malformed tool markup, and each skipped engagement remains visible in counters and receipts.*
[tests](#) [request API](#)

MEDIA

Image, audio and video request inputs

Media parts are decoded under explicit bounds and dispatched only to adapters with a multimodal hook; Gemma 3n and MiniCPM-o run an encoder prefill, then rejoin continuous decode. *Media-aware identity extends exact cache reuse without confusing different inputs or encoder boundaries.*
[decoding](#) [GPU smoke](#)

RESOURCES

Files, Batch and representation endpoints

Bounded tenant-scoped Files and JSONL Batch resources share the engine; embeddings and rerank require an adapter representation hook. *Client resource workflows remain useful without misrepresenting a generation model as a retrieval model.*
[resources](#) [parity](#)

ADAPTERS

Drained dynamic LoRA adapters

`/v1/load_lora_adapter` and `/v1/unload_lora_adapter` follow vLLM's shape; loading drains inflight work, checks exact tensor coverage, invalidates cached state and advances the model revision. *A fine-tune can be swapped in without a restart, and no prefix computed under other weights is reused.*
[install](#) [lifecycle](#)

TOOLS

Operator-allowlisted MCP tool backend

With `--tool-backend-config`, Responses mcp tools run server-side against allowlisted Streamable HTTP servers, with approval fixed to never and at most eight model/tool rounds. *A local model can call operator-approved tools in one request while client-supplied servers stay unreachable.*

[backend](#) [request API](#)

AGENT CLIENTS

Real Codex and Claude Code turns, conformance-tested

The two agent clients the lab actually runs are detected from headers they send themselves, and driven through their own multi-turn loops against a recorded transcript rather than a hand-written approximation. *Agent compatibility is evaluated through real multi-turn behavior, not just endpoint names.*

[GPU evidence](#) [API parity](#)

TOOL GRAMMAR

Strict alternation and streamed tool calls

Under the opt-in tool grammar, `auto` becomes a real language — free text, or a call block, or a JSON answer — and calls that have completed are sent as they finish instead of being buffered to the end of the response. *A constrained agent turn stops being all-or-nothing at the last token: the client sees each call as soon as it is well-formed, and the terminal contract still runs before the stream closes.*

[grammar](#) [API parity](#)

LEGAL REQUESTS

Schema-legal tool request handling

Optional descriptions and single-call bounds are normalized before or during generation, while terminal tool contracts remain explicit. *Clients get predictable handling of valid requests and explicit failures for unsatisfied contracts.*

[contract](#) [API parity](#)

OPERATOR ACCESS

Authenticated tenants and local agent launchers

Optional API-key and tenant authentication complement Host/Origin checks, persisted reasoning keys and configured client launchers. *Local agent workflows can preserve identity and reasoning continuity under an operator-controlled service boundary.*

[operation and clients](#) [server controls](#)

APCv2 and transactional state ownership · reuse · persistence · 26

DEFAULT NEW

Sidecar-ancestor hits on draft routes

Self-MTP and external-draft lookups walk to the deepest resident ancestor that carries a usable draft sidecar. *A plain checkpoint deeper on the same path no longer turns a usable sidecar hit into a full miss.*

[APCv2](#)

DEFAULT NEW

Warm continuations cut at cold boundaries

A warm request now slices its prefill where a cold prefill of the same prompt would, and receipts say so. *Exact reuse is also cold-bit-identical where the receipt reports slice_aligned.*

[placement](#)

OPT-IN NEW

Value-aware APCv2 retention

Evicts the entry with the least saved prefill per freed unique byte first, instead of rank then recency. *Shared tool-schema and document checkpoints can outlive one-off prompt boundaries.*

[APCv2](#)

DEFAULT NEW

Shared snapshots charged once

A physical-buffer ledger charges aliased recurrent and window snapshots to one resident holder. *The APCv2 byte cap keeps more entries; one repro charged 3.25 MiB for 2.25 MiB held, up to about 5x on long hybrid prompts.*

[APCv2](#)

ONE AUTHORITY

APCv2-only prefix reuse

A radix index locates frozen segments; there is no legacy cache or hidden second serving backend. *Every optimization shares one ownership, identity and eviction contract.*

[APCv2](#) [architecture](#)

ISOLATION

Layered copy-on-write branches

Borrowed branches retain leases and mutate private suffixes while immutable prefix state remains shared. *Warm reuse and fanout stop being synonyms for shared mutable state.*

[COW cache](#) [branch transaction](#)

IDENTITY

Revision-bound state manifests

Append, fork, rollback, restore, spill and compact advance explicit revisions across attention, recurrent, sparse, draft, RNG, grammar and transcript planes. *Stale work cannot overwrite a newer request state; approximate publication cannot masquerade as exact.*

[transactions](#) [contracts](#)

PAIRED STATE

Atomic target + draft snapshots

Prompt-boundary target cache and MTP sidecar publish and restore together; ordinary decode publishes an equivalent target boundary. *Prefix reuse remains correct across ordinary/speculative route changes and warm cohorts.*

[sidecars](#) [lifecycle tests](#)

FANOUT

Cache capsules for B1→Bn materialization

Generation-stamped capsules construct exact batch-ready plain-KV planes outside the scheduler, validate compatibility late and dispose stale products. Mixed caches combine capsule and ordinary planes in one reservation. *Parallel siblings reuse one committed boundary without turning asynchronous construction into an ownership race.*

[capsules](#) [campaign report](#)

DURABILITY

Checksummed persistent APC blocks

Optional immutable blocks use atomic manifests, strict contained paths, identity checks, corruption detection and orphan cleanup. *Large process-local snapshots can be restored and deduplicated with bounded blast radius.*

[block format](#) [tests](#)

TIERS

Resident cache plus bounded idle disk spill

Resident checkpoints are capacity-bound; idle exact snapshots spill to a process-local 64 GiB scratch tier and restore as misses when corrupt or incomplete. *Useful long prefixes can outlive memory pressure without pretending disk state is a durable conversation database.*

[cache policy](#) [implementation](#)

TOPOLOGY

Segmented shared-prefix QSA

A common immutable base and per-lane suffixes are attended separately and merged with exact online-softmax math. *Branching long-context cohorts avoid materializing a dense copy of the shared prefix for every row.*

[shared suffix](#) [Metal probe](#)

REPRESENTATION

Asynchronous physical QSA promotion

Eligible segmented cohorts may build a physical representation on a separate stream, then publish it as a validated successor transaction. *The runtime can construct an alternative representation while preserving the authoritative segmented state until validation.*

[promotion](#) [pressure review](#)

SURGERY

Spomin live KV surgery

Adapter-owned QSA or standard-attention surgery compacts eligible private ordinary caches while preserving an exact pre-surgery shadow. *Eligible long contexts can reclaim memory without publishing compacted state as exact.*

[surgery](#) [tests](#)

ROLLBACK

Rotating-cache undo and accepted-prefix replay

Multi-cache checkpoints roll back atomically, replay only accepted tokens and fail closed on stale lineage. *Sliding-window and recurrent layouts can participate in exact speculative transactions.*

[replay](#) [segmented ring](#)

LONG CONVERSATIONS

Revision-bound transcript compaction

Oldest, largest or lowest-importance immutable transcript segments can be selected while protecting system, recent and tool-paired turns; the uncompacted ledger remains a separate plane. *Context reduction becomes an explicit bounded-fidelity operation rather than silent history deletion.*

[compaction](#) [tests](#)

FIDELITY

Revision-bound Q8 and K8V4 cache operations

Named quantized-KV operations carry explicit approximate fidelity, adapter compatibility and evidence requirements. *Lower cache residency is available through an explicit fidelity contract, not silent quantization.*

[controller](#) [provenance](#)

HYBRID REUSE

Exact interior hybrid checkpoints

Budgeted exact target and draft snapshots inside a hybrid prompt let a diverging request resume recurrent state below the prompt end instead of re-prefilling from zero. *Edited or branched long prompts on GDN hybrids reuse the deepest exact state that fits measured headroom.*

[lattice and capture](#) [cache policy](#)

RESTARTS

Restart-surviving APC persistence

Exact APCv2 entries can live in an identity-bound directory that is rescanned from manifests at startup, with corrupt entries quarantined and every restore digest-verified. *Long prefixes survive a restart or suspend without trusting stale, foreign or tampered state.*

[persistence](#) [CPU qualification](#)

SESSIONS

Tenant-owned APC sessions

A session_id tags a tenant's APCv2 entries so clients can park them to disk with a TTL, resume them with an idle-boundary prefetch, inspect them and delete them. *Agents and long conversations keep their warm state across idle gaps under explicit per-tenant byte caps.*

[session lifecycle](#) [HTTP tests](#)

RECOVERY

Committed speculative recovery checkpoints

Prompt lookup, external drafts and segmented self-MTP share one generation-stamped checkpoint of the last committed boundary, restored only on a matching route, revision and position. *A failed speculative round falls back to proven state instead of a full re-prefill or a guess.*

[recovery slot](#) [tests](#)

CACHE HYGIENE

Tenant-scoped, write-suppressible prefix cache

An operator flag binds the APCv2 namespace to the tenant header, and a per-request flag stops a request from publishing any prefix state while still letting it read. *Authenticated tenant scope can isolate prefix reuse, while one-off requests can suppress cache publication.*

[namespace and write control](#) [cache controls](#)

INTERIOR REUSE

Interior checkpoint reuse with cold replay

Captured exact prompt boundaries let recurring system and document prefixes avoid repeated prefill, with cold replay as the correctness control. *Reusable prompt structure saves prefill while retaining a direct answer-equivalence check.*

[GPU evidence](#) [serving guide](#)

SEMANTIC DIRECTORY UPDATED

Revision-bound semantic capsules

A layered directory resolves model, tenant, session and request handles into immutable semantic capsules. *Durable memory has an explicit owner, scope and revision alongside the exact token cache.*

[directory](#) [capsules](#)

SEMANTIC RECALL

Classifier-gated memory after response delivery

Retrieved concepts and relationships enter a bounded semantic preamble; proposed facts pass confidence and margin gates before durable publication. *A useful memory can persist across turns without treating every generated sentence as established knowledge.*

[semantic memory](#) [serving middleware](#)

CONCEPT EXPERIMENTS

Neural concepts with explicit fidelity

Recurrent and deep concept bridges, concept-token steering and adaptive capsule experiments have independent source-bound policies. *Semantic retrieval and neural state modification can be evaluated without confusing either with exact prefix reuse.*

[neural concepts](#) [bridge receipts](#)

Scheduling and memory control throughput · fairness · survival · 22

RESEARCH NEW

Adaptive batch geometry and packed varlen prefill

A planner chooses rectangular, bucketed, packed or mixed geometry; packed rows remove padding from MLP and prefill work. *Skewed cohorts stop paying for padding rows: 1.52x at 39.8% padding removed in the N3 measurement.*

[planner](#)

OPT-IN NEW

GPU keep-warm ticker

A one-element kernel every 0.5 s keeps the GPU out of its idle power state for N seconds after real work. *Agent tool rounds can skip the idle-wake stall; on an M5 Max it measured about 0.3 s after 20 s idle, so it stays opt-in.*

[worker](#)

DEFAULT NEW

Co-arrival hold for self-MTP cohorts

Multi-chunk prompts that arrive together now form one segmented MTP cohort instead of locking it at width 1. *Concurrent chat prompts over one prefill step decode in a single cohort rather than one MTP lane plus plain lanes.*

[scheduler](#)

NEW DEFAULT NEW

Completion-time stall estimator

Prefill slices beside live decode are sized from $t = a + b \cdot R$ per route kind and KV depth instead of a running maximum. *One fast sample can no longer license a slice that stalls neighbours for seconds after a slowdown.*

[policy](#)

CONCURRENCY

Continuous batching with truthful widths

Requests join and leave active batches independently; receipts report physical compute widths rather than equating HTTP concurrency with batched execution. *Throughput claims and capacity planning are tied to what the model actually executed.*

[batch engine](#) [metrics](#)

PREFILL UPDATED

Adaptive, bounded prefill

Chunk sizes respond to prompt length, workload and contention while remaining inside configured and measured limits; prompt-length autoscaling picks 512, 2048 or 8192 rows when neither operator nor adapter sets a step. *Long prompts stop monopolizing the generation loop.*

[admission](#) [policy](#)

FAIRNESS UPDATED

Wall-time decode debt

Contention-aware prefill caps measure stall cost and repay it with decode time; since this sweep the stall bound comes from a completion-time cost model per route kind and KV depth. *Interactive generations keep making progress beside expensive prompt ingestion.*

[fairness](#) [source assessment](#)

ADMISSION

Physical memory accounting

Host availability, Darwin process footprint, MLX active/cached bytes, model cache geometry, activation/transient reserves and retained state are reconciled before allocation. *Unified memory is treated as a shared operating budget—not “free RAM” until the machine swaps.*

[policy](#) [host metrics](#)

HEADROOM

Advisory-scaled service and driver reserve

Admission protects bounded service and driver headroom calibrated to Metal's working-set advisory. *Memory admission scales across Macs while preserving operating headroom.*

[geometry](#) [memory analysis](#)

RECOVERY

Deferred memory admission without lane capture

Waiting requests retain bounded leases, retry measured recovery every 250 ms and expire explicitly, while active work continues. *Temporary allocator lag does not waste an execution lane or force an over-budget request to run.*

[tests](#) [semantics](#)

RECLAMATION

Optional work yields before useful state

Promotion is settled, canceled and drained before checkpoint eviction; accounting is remeasured and never credits hypothetical freed bytes. *An optimization cannot destroy the warm prefix it was meant to accelerate.*

[review](#) [tests](#)

COHORTS

Atomic multi-lane admission and uniform fallback

Tenant/cohort identity, whole-cohort staging, uniform lower draft depth and transactional rollback prevent partial-width execution. When the contract cannot be met, the cohort fails together. *Batch semantics remain predictable under pressure instead of silently degrading lane by lane.*

[cohort tests](#) [controller](#)

EXTERNAL DRAFT

Partial-fit fair progress

External-draft lanes rotate selection, advance a fitting subset before reclaiming and fall back to ordinary only for recoverable drafter failure. *One expensive lane does not starve smaller ready work, and cache state remains protected.*

[executor](#) [adapter](#)

LIVENESS

Dead-worker and stall failure semantics

A dead generation thread fails readiness and exits for supervision; requests unable to progress for 60 seconds terminate explicitly. *The service cannot look healthy while generation has silently died.*

[service](#) [contract tests](#)

LIFECYCLE

Drain, suspend and resume in place

Admin quiesce closes admission, lets accepted work finish, and can spill every exact warm checkpoint to the APC disk tier. Resume reopens with weights still loaded and caches restored lazily or by prefetch. *Operators can take the server out of rotation and free memory without killing the process, reloading the model or losing warm prefixes.*

[quiesce lifecycle](#) [quiesce semantics](#)

LATE ARRIVALS

Short prompts overtake long prefills

A request that fits in one prefill chunk alternates with, or runs beside, long chunked prompts instead of queueing behind them. A lane held by a segmented width lock falls back to ordinary decode. *Short interactive requests get a first token within a few scheduler rounds even while a long document is being ingested.*

[starvation tests](#) [probe record](#)

ALLOCATOR

Speculative decode returns freed buffers

The self-MTP loop clears MLX's freed-buffer pool whenever emitted tokens cross a 256-token boundary. Prompt lookup clears it after each prefill chunk and on the shared decode cadence. *Multi-token decoding no longer builds a hidden cache of dead KV buffers that can reach twice the working set.*

[27B 64K evidence](#) [tests](#)

ADMISSION

Admission that avoids false 429s

Parallel samples, warm hits and optional checkpoints are charged at measured cost. Admission waits or reclaims allocator memory before it refuses. *Measured accounting and bounded recovery reduce avoidable refusals while retaining real memory limits.*

[admission](#) [projection tests](#)

QUEUE ORDER

Shortest-remaining-prefill scheduling

Native MTP prioritizes shorter remaining prefills with a bounded bypass cap that preserves progress for long requests. *Short interactive requests can make progress without starving larger jobs.*

[scheduler](#) [GPU evidence](#)

RESIDENCY

MoE experts paged from disk

Individual MoE expert rows are read from the safetensors file on demand into a bounded resident cache instead of being held whole, behind a default-off policy. *A bounded memory footprint becomes possible where full expert residency is impractical, at an explicit I/O cost.*

[streaming](#) [bit-identity gate](#)

LOAD PEAK

Per-shard unified-buffer-cache eviction

Each checkpoint shard is materialized and its unified-buffer-cache mirror evicted before the next is read, so the mirror is bounded by the largest shard rather than the whole model. *Lower duplicate file-cache residency leaves more headroom for the model, its drafter and request state during startup.*

[shard eviction](#) [tests](#)

OPT-IN **NEW**

Power governor with a runtime watt budget

`--power-governor max_throughput`, efficient or budget, steered by measured GPU + DRAM watts; `/v1/admin/power` changes mode and budget at runtime. *An operator or a grid demand-response signal can cap the server's draw without restarting it: a 25 W budget held 24.5 W on Qwen3.8-27B.*

[governor](#) [operator guide](#)

Speculation and Flash-Next acceleration [draft](#) · [verify](#) · [hybrid kernels](#) · 21

QUALIFIED, UNSELECTED **NEW**

QSA direct4 / direct8 selectors

Exact alternatives to the radix selector remove about 1 GiB of selector scratch at 8192 x 32768. *Long-context Flash-Next can drop a large transient without changing selected blocks.*

[selector](#)

OPT-IN **NEW**

Exact block verification for linear external drafts

Sun et al. block verification accepts a never-shorter prefix in expectation while keeping the target law, on the linear DFlash2 chain. *Sampled drafts can keep more tokens per round without approximate acceptance.*

[route](#)

DEFAULT ROUTE **NEW**

Qwen3.8 DFlash2 tree with an owned TensorFold executor

One service verifies a 15-node draft tree at one lane, 7/4/3 nodes per lane at two to four lanes and a capped chain above, in about 16 target rows per forward. *The pinned 27B pair decodes 1.5x the ordinary aggregate at four lanes in the sweep A/B, with the executor vendored in mlx2 rather than imported.*

[policy](#)[executor](#)

NATIVE SPECULATION

Persistent self-MTP with ordinary handoff

Embedded draft-head state persists across rounds; eligible wide cohorts can transfer committed state to ordinary decode in a one-way handoff. *Single-stream speculation and wide-batch ordinary decode can share one state-safe lifecycle.*

[segmented MTP](#) [executor](#)

ADAPTIVE SPECULATION

Cohort-wide adaptive MTP depth

One depth decision per cohort responds to acceptance at closed boundaries, including exact target-only depth zero and deterministic probes. *The runtime can stop paying for unproductive drafts and test when speculation becomes useful again.*

[controller](#) [tests](#)

DRAFT-FREE **UPDATED**

Prompt-lookup decoding

An explicit candidate route proposes repeated text from bounded indexed lookup, then verifies and commits it against the target; hybrid recurrent targets now verify all lanes in one batched forward. *Copy-rich workloads can explore draft-free acceleration while retaining ordinary decode as the reference.*

[route](#) [PLD core](#)

EXTERNAL DRAFT

Muse DFlash2 linear speculation

A separate drafter proposes linear blocks; the target verifies with exact residual sampling and per-lane rotating-cache rollback. *The product supports a second speculative family without contaminating native-MTP contracts.*

[port report](#) [drafter](#)

CORRECTNESS

Exact speculative sampling law

Accepted proposals, rejection corrections and bonus tokens use transformed target probabilities with explicit rollback and synchronization. *Acceleration does not quietly substitute the draft model's distribution for the target model's contract.*

[sampling](#) [sync](#)

PLE

File-backed learned n-gram embeddings

PLE rows live outside the main checkpoint in a validated file, use bounded LRU residency, compiled lookup and known-tail prefetch. *Large learned prompt features become a demand-paged serving resource instead of permanent resident weight.*

[PLE store](#) [prefetch tests](#)

GDN

Fused recurrent decode, verify and compact replay

Gated DeltaNet paths fuse decode/verification work and can replay compact rollback state, with counter-gated fallbacks. *Hybrid recurrent layers participate in speculation without full state copies on every rejected suffix.*

[fused GDN](#) [GPU qualification](#)

QSA

Pooled, indexed and private-delta attention

Pooled keys, scatter-chosen access, indexed long-context kernels and private suffix deltas are selected by source, shape, context and output-budget gates. *Flash-Next's sparse attention topology is optimized directly instead of flattened into generic dense attention.*

[indexed QSA](#) [mechanism matrix](#)

QSA VARIANTS

NAX, split-stage, exact-set folding and native epilogues

Alternative prefill/decode paths, two-row exact-set folding, fused output-gate and sequential merge components retain independent gates and oracles. *Kernel innovation can be selected per geometry rather than promoted as one monolithic "fast mode."*

[NAX](#) [exact-set gate](#)

MOE

Fused MoE gate/up and expert dispatch

Gate/up projection and scalar/tile expert kernels use explicit shape gates with stock fallbacks and separate engagement counters. *Sparse expert models get Apple-specific fast paths without model-name branches in the scheduler.*

[fused MoE](#) [router](#)

SELECTION DISCIPLINE

Ordinary reference route beside every optimized route

The same model, APCv2 and request lifecycle can run without speculation, preserving a clean comparison and recovery path. *Optimization is measurable, reversible and never the only way to serve a model.*

[reference contract](#) [boundary tests](#)

VERIFY KERNELS

Batched-verify quantized matmul kernels

Speculative verify matmuls fold strided lanes into one M problem, tile up to eight vectors and, at M=8-16 on M5, dequantize into threadgroup memory for neural-accelerator matmul2d. *Eligible verify rows share weight reads, with numerical and performance behavior bound to the native runtime and geometry.*

[NAX small-M evidence](#) [strided-verify evidence](#)

DRAFT FIDELITY

Byte-keyed MTP norm repair

At load, the Qwen3.6-35B adapter adds the missing +1 to MTP-head RMSNorm gammas that a converter left unshifted, matching each tensor by the SHA-256 of its stored bytes. *Recognized conversion faults can be corrected without applying a risky value-based heuristic to unrelated artifacts.*

[norm repair](#) [GPU A/B](#)

M5 PREFILL

Instance-scoped int8 NAX prefill

Opt-in W8A8 prefill routes prompt-sized projections of one model through int8 matmul2d on M5 neural accelerators, with its prefix state in a separate APCv2 namespace. *Long prompts can use M5 int8 throughput without changing decode, verification or any other model in the process.*

[int8 prefill](#) [Xing4.0 evidence](#)

RELAXED VERIFY

Opt-in FLy relaxed verification

An execution-policy-only option accepts an isolated high-entropy rejection when the next window of draft tokens all pass ordinary verification. *Operators can trade exactness for longer accepted runs on drafts that briefly disagree, with every relaxed accept recorded in receipts.*

[FLy policy](#) [tests](#)

PROPOSAL POLICY

Exact B1 copy drafts and independent verify gates

Flash-Next and Qwen3.8 enable exact copy proposals at one lane; wider cohorts and bit-exact verify retain their own eligibility and evidence boundaries. *A bounded, copy-friendly workload can avoid draft-head work without making the same promise for every cohort.*

[bit-exact evidence](#) [copy-draft evidence](#)

QUANTIZED DECODE

GQA-aware Q8 and K8V4 attention

Quantized cache kernels fold grouped query heads and use tiled Metal decode where their geometry is supported. *Eligible long-context requests can reduce cache traffic and residency while reporting approximate fidelity honestly.*

[fidelity gate](#)

DEFAULT **NEW**

6-bit affine NAX sorted gather

The fused MoE sorted gather admits 6-bit affine experts (widths 4, 6, 8). *Models with 6-bit expert layers keep the fast gather: Qwen3.6-35B-A3B 8K TTFT about 7% lower, bit-identical.*

[provenance](#)

Model-family vertical slices [adapters](#) · [topology](#) · [strict loading](#) · 16

CLOSED **NEW**

LLaDA direct denoising lane

Active-block postprocess and active-head projection for LLaDA reached exact component parity. *A diffusion LM family is measured on the same discipline, and the negative result is kept.*

[adapter](#)

FLAGSHIP

Qwen4 / Qwen3.8 Flash-Next

Mixed quantization, embedded MTP, 12 QSA target layers, recurrent GDN layers, PLE and MoE are served as one topology-aware stack. *The release targets the frontier hybrid architecture rather than limiting Apple serving to conventional dense transformers.*

[model](#) [parity map](#)

DENSE HYBRID **UPDATED**

Qwen3.8 27B

Strict artifact binding, dense hybrid adapter, segmented plain-KV execution, native MTP, and, for the pinned target and drafter revisions, an external DFlash2 tree route selected by default. *The same state and qualification contracts extend beyond the flagship artifact.*

[port report](#) [model](#)

MOE HYBRID

Qwen3.6 35B-A3B

GDN/MoE ordinary and native-MTP artifact slices, APCv2, bounded fused-GDN and prompt-lookup integration. *The same cache and scheduler contracts support hybrid MoE while preserving artifact-specific behavior.*

[port report](#) [model](#)

ALTERNATIVE ARCHITECTURE

Muse Glimmer + DFlash2

Ordinary text generation, model-specific reasoning controls and an external linear drafter use global/windowed rotating caches and exact verification. *The runtime demonstrates architecture breadth without forcing Qwen-specific math into common scheduling.*

[Glimmer](#) [DFlash2](#)

CODE MODEL

North Mini Code

An ordinary-serving slice covers model configuration, memory geometry, output conventions and qualification boundaries for a code-specialized family. *Code-model serving remains source-grounded, with reasoning and tool behavior owned by its adapter.*

[port report](#) [adapter](#)

LOADING

Strict artifact and tensor reconciliation

Model adapters validate configuration, topology, tensor inventory, safetensors headers, mixed quantization and auxiliary manifests before allocation. *A similarly named but incompatible checkpoint cannot accidentally authorize the wrong execution path.*

[artifact identity](#) [ledger](#)

MLA + MHC

Xing4.0-29B-A4B with native self-MTP

DeepSeek-style MLA attention, a noaux sigmoid MoE and four-stream manifold hyper-connections are served with fused Metal mHC kernels and the checkpoint's embedded MTP layer. *MLA and multi-stream residual math reuse shared state and scheduling contracts through a strict adapter.*

[port report](#) [mHC kernels](#)

SLIDING MOE

Laguna XS 2.1 ordinary serving

poolside's 40-layer sliding/global sparse MoE is served through APCv2, continuous batching, prompt lookup, grammar and its XML-style tool parser, with vendor sampling defaults. *Another vendor's agentic coding model joins the shared serving stack, and experimental kernels are kept behind measured gates.*

[adapter](#) [provenance](#)

MEDIA INPUT

Gemma 3n and MiniCPM-o media paths

Image, video-as-frames and input-audio requests run an isolated encoder prefill through mlx-vlm, then rejoin ordinary continuous decode with media-fingerprinted APCv2 reuse. *Multimodal models are served without a second server, and a cached text prefix cannot be reused across different media.*

[mlx-vlm review](#) [media policies](#)

ORDINARY TOPOLOGY

Broader strict ordinary-model adapters

Registry dispatch covers dense, MoE, sliding-window and hybrid families through topology-specific adapters. *A model family can enter the ordinary lifecycle before speculative or media capabilities are ready.*

[registry](#)

GEMMA 4

Gemma 4 text, image and video candidates

26B A4B and 31B adapters accept retained BF16 and affine 8-bit layouts with mlx2 full and rotating KV state. *Dense and sparse Gemma models can reuse the same exact cache and request lifecycle around model-owned media encoders.*

[adapter scope](#)[implementation](#)

NEMOTRON 3 SUPER

Hybrid Mamba, MoE and attention

Nemotron 3 Super 120B-A12B has a strict 5-bit adapter with an ordinary default and bounded embedded-MTP experiments. *A different recurrent architecture is represented explicitly instead of forced into a plain-KV assumption.*

[model and evidence](#)[adapter](#)

MEDIA CACHE BRIDGES UPDATED

LFM2.5-VL, SmoVLM2 and Qwen2.5-VL candidates

Source-backed bridges preserve each family's media boundary and state topology through APCv2. *Repeated media prompts can reuse exact committed state without losing recurrent planes or positional metadata.*

[registry](#) [admission](#)

DIRECT GENERATION

Qwen-Image 2.1 and LTX 2.5 direct adapters

Image generation/editing and audio/video diffusion use direct media adapters with pinned conversion and loader provenance. *Non-causal media models can be explored without inventing a token prefill/decode contract they do not have.*

OFFLINE DIARIZATION

Nemotron 3 speaker-turn CLI

A separate mlx2-diarize command processes 16 kHz mono PCM WAV into speaker turns under a narrowly qualified offline profile. *Speaker attribution gains a source-bound evaluation without being confused with text generation or live transcription.*

Assurance, observability and operational knowledge proof · failure · learning · 17

RULE NEW

Near-tie qualification rule

A divergence where both top-two margins are at most 0.5 nat on a shared prefix, with the functional oracle passing, is classified a near tie rather than a failure. *Batched and speculative routes are judged on real defects, not on BF16 reduction-order coin flips.*

[procedure](#)

TOOLING NEW

Feature profiler and thermal map

`profile_features` drives real servers per arm through HTTP; `thermal_map` runs all arms at equal heat from cold through sustained and duty-cycled load. *Route comparisons carry receipts, counter deltas, swap, energy per token and output identity, and heat is controlled rather than ignored.*

[profiler](#) [thermal map](#)

OPT-IN NEW

Power and thermal telemetry from the server

`--power-telemetry` publishes per-subsystem watts, joules, GPU P-state residency, die temperatures and a rolling J/token without root. *Energy per token becomes a first-class serving metric next to tokens per second.*

[telemetry](#)

ROUTING UPDATED

Explicit route confidence and capability gates

Qualified, unqualified and candidate startup modes expose their confidence honestly while protected operations retain specific evidence gates. *Users can distinguish runnable software from a tested profile and from what actually executed.*

[planner](#) [gate](#)

IDENTITY

Source, runtime, artifact and policy binding

Receipts bind source hashes, native MLX, dependencies, tokenizer/config/manifests, artifact metadata and serving settings to an independently pinned qualifier. *A passing result cannot be transplanted onto a different binary, model or policy.*

[qualification](#) [experiments](#)

OBSERVED USE

Mechanism counters and route receipts

Responses and status expose route identity, cache reuse, MTP acceptance, actual widths, structured-output bounds and kernel engagement/decline reasons. *Operators can distinguish enabled, eligible, selected and executed.*

[telemetry](#) [required counters](#)

STATUS

Health and batching introspection

`/health`, `/v1/status` and `/v1/status/batching` expose readiness, memory, APC leases, queue depth, scheduler decisions and policy identity. *A supervisor and an operator see the same execution truth.*

[endpoints](#) [tests](#)

FAILURE MODEL

Fail-closed capability boundaries

Missing capabilities, stale lineage, unsupported schemas, incomplete cohorts and mismatched evidence are explicit errors or recorded declines. *The system refuses unsafe degradation that would make a benchmark look successful.*

[design](#) [regressions](#)

SECURITY POSTURE

Hostile-request hardening

Input envelopes, output limits, finite numeric checks, vocabulary bounds, bounded stop lists, contained cache paths and strict JSON parsing constrain abuse. *The local server has explicit resource and parser boundaries even before an authenticated front end.*

[limits](#) [validation](#)

PROVENANCE

Mechanism-by-mechanism provenance

Mined mechanisms carry source, license and validation records; optional multimodal dependencies are declared explicitly. *A small core remains auditable even when it draws on a wider model ecosystem.*

[ledger](#) [notice](#)

QUALIFICATION

Functional, adversarial and mechanism gates

Cold/warm HTTP, stream, tools, reasoning, sampling, logprobs, batching, mixed prefixes, context, cancellation, recovery, strict schema, state tests and lease drainage form one acceptance envelope. *Qualification measures the assembled product, not an isolated kernel.*

[qualifier](#) [method](#)

NEGATIVE KNOWLEDGE UPDATED

Unprofitable paths are closed, not forgotten

ASPIRE-style scheduling, whole-decode compilation, megakernels, certain fused merges and other candidates retain evidence explaining why they are not automatic defaults. *The project compounds learning and avoids repeatedly “discovering” attractive but workload-negative optimizations.*

[ASPIRE](#) [closeout](#) [excluded paths](#)

TELEMETRY

Prometheus metrics and OTLP traces

GET /metrics serves a versioned, label-bounded Prometheus schema covering requests, latency histograms, lifecycle, APCv2 and every advanced mechanism, and optional OTLP/HTTP spans carry W3C trace context. *Standard dashboards and alerts can observe what executed without a scrape touching the device or leaking prompt data.*

[metric contract](#) [exporter](#)

CAMPAIGNS

Resumable route qualification campaigns

Pinned campaigns combine functionality, client compatibility, mechanism observations, graded workloads and context regression. *Qualification work is reproducible and restartable, with incomplete evidence clearly bounded.*

[campaign](#) [20x20 results](#)

PEER PROBES

Peer bugs become regression tests

Selected failure modes from surveyed peer-engine PRs are reproduced against mlx2, judged impacted or not with evidence, and kept as CPU regression or mechanism-assertion tests. *Concrete failure modes become durable local checks with explicit state and mechanism expectations.*

[09-19 probes](#) [mechanism assertion](#)

HOST TELEMETRY UPDATED

mlx2-top live monitor with unprivileged power

A terminal display shows live inference metrics, memory, swap and route receipts, and now reads SoC power, GPU P-state residency and die temperatures through IOReport without root. *Operators see resource pressure, energy and observed mechanism engagement during long tests without a sudo prompt.*

[terminal UI](#) [collector](#) [tests](#)

REPRODUCIBILITY

Prefill geometry in every receipt

Actual chunk sizes and execution widths record a request’s numerical path and help explain differences across schedules. *Scheduling choices become part of the evidence needed to reproduce an answer.*

[GPU evidence](#) [receipt schema](#)

Deep local state, in a shared field

Carried over from the September edition and still true, with this week's additions.

Apple-native servers and larger inference engines overlap on APIs, batching, prefix reuse and speculation. mlx2's focus is the composition: adapter-owned model state, a single exact prefix-cache authority, physical memory accounting and receipts that distinguish configuration, confidence and execution.

Apple-native serving · oMLX, Rapid-MLX. oMLX documents continuous batching, shared copy-on-write prefixes and memory/SSD cache tiers. Rapid-MLX combines an MLX server and Mac application with OpenAI/Anthropic APIs, coding-agent tools and release gates. Overlapping capabilities; a cache or endpoint alone is not a uniqueness claim.

Native speculative decoding · MTPLX. MTPLX specializes in native MTP on Apple silicon. mlx2 combines embedded-head MTP, prompt lookup and external drafting (now a token tree) with ordinary references, adaptive depth, memory admission and cache transactions.

Desktop and client workflows · Ollama, LM Studio. Ollama provides a documented subset of OpenAI-compatible APIs; LM Studio documents Responses, Chat Completions, Embeddings, tools, structured output and Anthropic Messages. mlx2's shared lifecycle serves familiar clients and refuses unsupported features explicitly.

Broader engines · vLLM, llama.cpp. vLLM documents automatic prefix caching including hybrid-state considerations; llama.cpp documents parallel decoding, continuous batching and server APIs. mlx2 concentrates its state and evidence contracts on Apple silicon instead of claiming wider coverage.

Capability comparisons from primary documentation checked on 27 September 2026, not benchmark rankings. This week's peer PR survey (oMLX, Rapid-MLX, llama.cpp, vLLM, SGLang, MTPLX, mlx-vlm and others since 22 Sep) found most peer mechanisms already present and fed four applicable peer bugs into the sweep (B1–B4).

The moat is the composition

A single state constitution

APCv2, request revisions, fidelity and plane identities cover warm reuse, speculation, rollback, spill, surgery, compaction and bounded approximation.

Hybrid architecture depth

PLE, QSA, GDN, MoE, MTP and now draft trees are optimized as interacting parts of one model, with their memory geometry and rollback semantics.

Receipts all the way down

The API result connects to scheduler decisions, cache reuse, actual width, speculative acceptance, kernel counters, and now energy and the power governor's actions.

Failure as a product feature

Partial fanout, unsupported fidelity, stale revisions, unobserved mechanisms and memory shortfalls are designed outcomes.

Provenance without dependency drag

Proven ideas are mined mechanism by mechanism and attributed; the TensorFlow executor is vendored and owned rather than imported.

Institutional memory

Negative experiments (this week: N20 at arbitrary traffic, GDN chunk-16, LLaDA, a histogram MoE planner) keep folklore from becoming defaults.

READ THE EVIDENCE AT ITS ACTUAL SCOPE

- **Confidence:** a route may run unqualified with an explicit label; only matching source, runtime, artifact and settings evidence supports “qualified”. The new Qwen3.8 default is a candidate.
- **Execution:** HTTP concurrency, configured lanes and physical model width are different measurements; receipts report observed work.
- **Fidelity:** exact reuse, numerically bounded kernels and approximate operations carry different contracts; near ties are classified, not ignored.
- **Performance:** sweep A/Bs are two-repetition evidence on one laptop; quoted performance needs a controlled run, and heat moves results.
- **Maturity:** mlx2 is experimental research software; interfaces, formats and policies are still moving.

EVIDENCE TRAIL

Built from source, checked against source

Every number above names its file; these are the files.

1. `qualification/runs/sweep-20261006/{q38-routes,fn,q38-features,q36,thermal-q38}/` — before campaigns
2. `qualification/runs/sweep-20261006/{estimator,keepwarm,telemetry-overhead,governor,post-q38-routes,post-fn}/` — same-code A/B or intermediate campaigns
3. `qualification/runs/sweep-20261006/{final-q38-routes,final-fn,final-q36-6bit,final-governor,dflash-comp}/` — final campaigns
4. `qualification/runs/sweep-20261006/lanes/adapters-other.md` — Other adapters lane report
5. `qualification/runs/sweep-20261006/lanes/adapters-qwen.md` — Qwen adapters lane report
6. `qualification/runs/sweep-20261006/lanes/apc-state.md` — APCv2 & state lane report
7. `qualification/runs/sweep-20261006/lanes/capsules.md` — Capsules lane report
8. `qualification/runs/sweep-20261006/lanes/config-qual.md` — Config & qualifier lane report
9. `qualification/runs/sweep-20261006/lanes/kernels-rows.md` — Kernels & rows lane report
10. `qualification/runs/sweep-20261006/lanes/pr-survey.md` — Peer PR survey lane report
11. `qualification/runs/sweep-20261006/lanes/serving-sched.md` — Serving & scheduler lane report
12. `qualification/runs/sweep-20261006/lanes/spec.md` — Speculation lane report
13. `qualification/runs/sweep-20261006/lanes/history.md` — feature inventory 29 Sep – 6 Oct
14. `qualification/runs/sweep-20261006/FINDINGS-GPU.md` — coordinator's GPU ledger
15. `qualification/runs/research-20261006/` — MoE tile headroom, A8 Amdahl, GDN prefill
16. `qualification/policies/qwen38-27b-dflash2-tensorfold-unified.json` — default tree policy
17. `docs/SERVING.md`, `docs/ARCHITECTURE.md`, `docs/METRICS.md`, `docs/QUALIFICATION.md`, `docs/PROVENANCE.md`
18. `docs/research/semantic-activation-capsule-bus-2026-10-06.md`, `docs/research/2026-10-04-agent-handover.md`
19. MLX2 Technology & Research Review, 6 October 2026 (external, source pin 008eb011f)
20. git: first-parent and non-merge counts on 187c6aa23 since 29 Sep; sweep commits 187c6aa23..43d561e38

Rebuild: `python docs/brag-book/build.py --pdf`. Every chart and number is recomputed from the campaign files; finding counts follow the lane reports. Built 2026-10-07 from main @ 43d561e38 (GitHub pierre427/mlx2 8359344 (13 signed topic commits) · private main 43d561e38).